



GE Medical Systems

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Revision 00

GE Medical Systems

LightSpeed¹⁶ DAS & Detector Subsystem Troubleshooting

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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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14 July 1993

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Explosion**

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**ATTENTION
Danger
d'Explosion**

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

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Revision History

Revision	Date	Reason for change
0	12/05/02	Initial Release of Manual

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Chapter 1

Architecture and Support Tools

Purpose: This document contains information to assist GE Personnel in identifying and resolving issues related to the LightSpeed 16 Slice DAS and Detector subsystem. Detailed procedures and concepts are not addressed. These items can be found in the appropriate advanced System Service manual.

Section 1.0

Sub-System Level Block Diagram

Figure 1-1 is a top level diagram of the LightSpeed 16 DAS/Detector subsystem. Refer to the Advanced System Service Manual for detailed information on the cabling pinouts and signal descriptions.

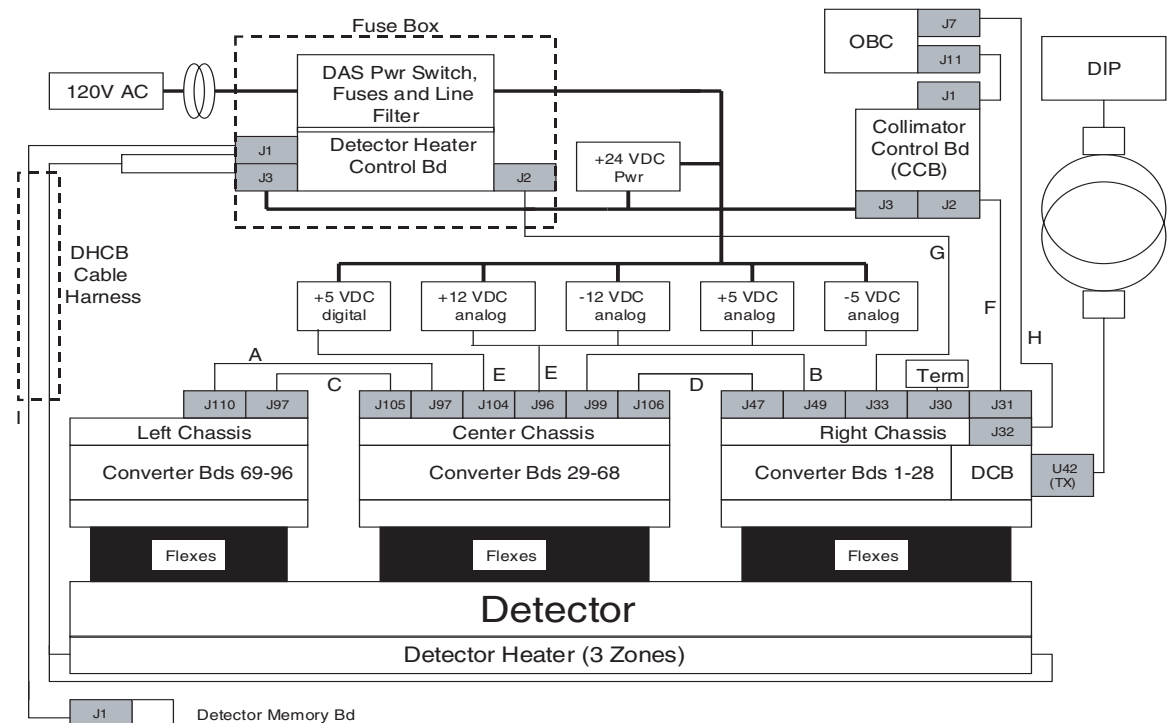


Figure 1-1 MDAS 16 Block Diagram

Section 2.0

Detector FET Mode and Interconnect Tables

[Figure 1-2 on page 9](#) and [Figure 1-3 on page 10](#) convey several important concepts.

- Detector has 24 rows but only 16 maximum can be active.
- Inner rows are 0.625 mm thick in Z. Outer rows are 1.25 mm thick in Z.
Inner rows are thinner and outer rows are thicker as shown in the referenced figures.
- Macro rows are identified using numbers.
 - Number 1 = macro row 1A and 1B = raw scan slice thickness.
 - Color coding included for visual separation.
- Z-Tracking channels collect row data that is not used.
 - This detail does not change the fault isolation process. It is mentioned only to state that this is an unimportant fact. Do not let it confuse you. These are identified as crossed out cells in the referenced figures.
 - These Z-Tracking FET modes were chosen as the most efficient for Z-Tracking control.

ULTRA 16 APPLICATIONS FET MODE SELECTIONS - 16 SLICE MODE																																
	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	Slice mode	FET mode	FET 1 LED	FET 2 LED	FET 3 LED	FET 4 LED	FET 5 LED	FET 6 LED
	2	1									4	3	3	3	2	2	2	2	2	2		1	2		Z-Chan	49	OFF	ON	ON	ON	OFF	OFF
	8	7											1	1	2	2	3	3	4	4	5	6	7	8	16x1.25	54	ON	OFF	OFF	ON	OFF	OFF
	2	1									4	3	3	3	2	2	2	2	2	2		1	2		Z-Chan	49	OFF	ON	ON	ON	OFF	OFF
	4	4											1	1	1	1	2	2	2	2	3	4	4		8x2.5	1	OFF	ON	ON	ON	ON	ON
	3	2	1								4	4	4	4	3	2	2	2	2	2	1	2	3		Z-Chan	48	ON	ON	ON	ON	OFF	OFF
			2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2			4x3.75	7	OFF	OFF	OFF	ON	ON	ON
				2	1	1													1	1	2				Z-Chan	39	OFF	OFF	OFF	ON	ON	OFF
					8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8					16x0.625	53	ON	OFF	OFF	ON	ON	ON
					2	1	1												1	1	2				Z-Chan	39	OFF	OFF	OFF	ON	ON	OFF
					4	4	3	3	2	2	1	1	1	1	2	2	3	3	4	4					8x1.25	0	ON	ON	ON	ON	ON	ON
							2	2	2	1	1	1	1	1	1	1	2	2	2	2	3	4	4		Z-Chan	1	OFF	ON	ON	ON	ON	ON
									4	3	2	1	1	2	3	4									8x0.625	3	OFF	ON	ON	ON	ON	ON

Figure 1-2 Detector Architecture and Application FET Modes

ULTRA 16 XRAY VERIFICATION FET MODE SELECTIONS - 16 & 8 SLICE MODE																																
A12 D12	A11 D11	A10 D10	A9 D9	A8 D8	A7 D7	A6 D6	A5 D5	A4 D4	A3 D3	A2 D2	A1 D1	B1 D1	B2 D2	B3 D3	B4 D4	B5 D5	B6 D6	B7 D7	B8 D8	B9 D9	B10 D10	B11 D11	B12 D12	Slice mode	FET mode	FET 1 LED	FET 2 LED	FET 3 LED	FET 4 LED	FET 5 LED	FET 6 LED	
									3	2	1	1	2	3										CAL 1	4	ON	OFF	ON	ON	ON	ON	
									3	2	1	1	2	3										CAL 1	4	ON	OFF	ON	ON	ON	ON	

Figure 1-3 Detector Architecture and DASTOOLS Interconnect FET Modes

Section 3.0

FET Control Interconnect

FET Mode control is achieved via six control lines.

Negative logic is employed, - 5 vdc = active.

Figure 1-2, Figure 1-3, Figure 1-1, Figure 1-4, and Figure 1-5 speak to FET Control LEDS and signal logic. FET control lines are active low or negative logic. This means DAS backplane LEDS are OFF when control lines are active.

There are three (3) sets of six (6) FET control lines driven by the **DCB**. These FET control lines form a binary value that gets decoded in the Detector and finally controls Detector Diode selection. The three (3) sets of FET Control are described in Table 1-1.

FET	DESCRIPTION	CHANNELS
Low Z	FET control lines for low z-reference channels	DAS Channels 1 - 8
Data	FET control lines for data channels	DAS Channels 9 - 762
High Z	FET control lines for z-reference channels. [Z-Axis tracking.]	High Z-Ref Channels 763 - 768

Table 1-1 FET Control Matrix

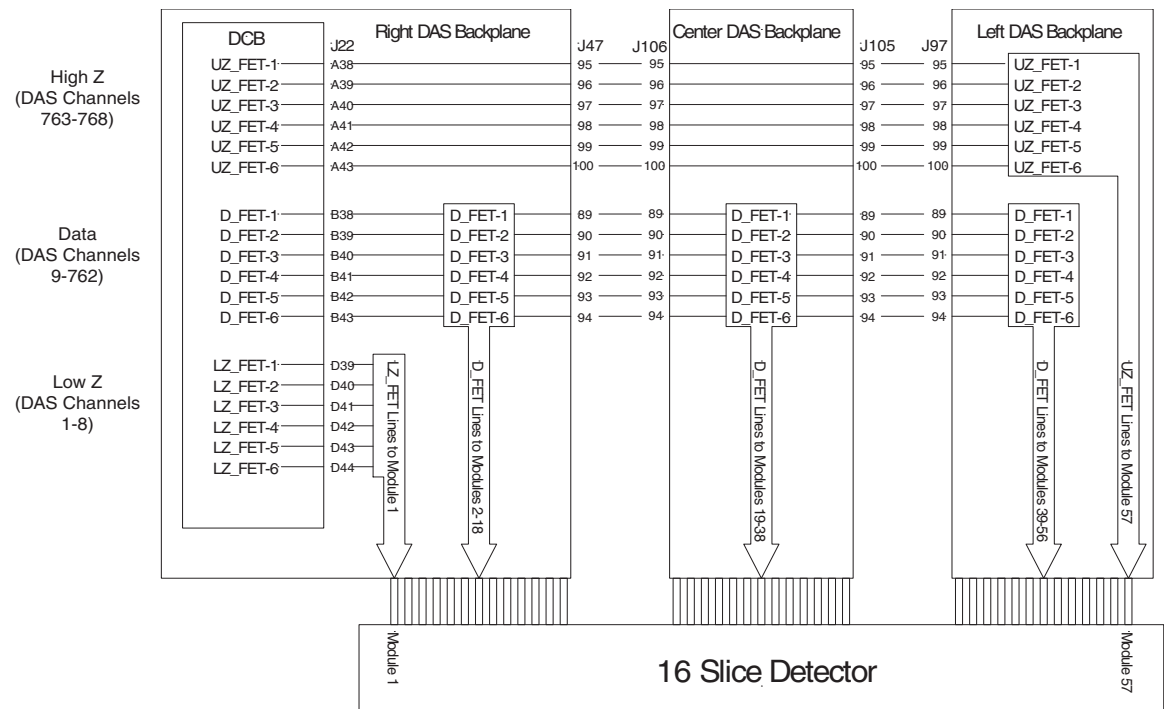
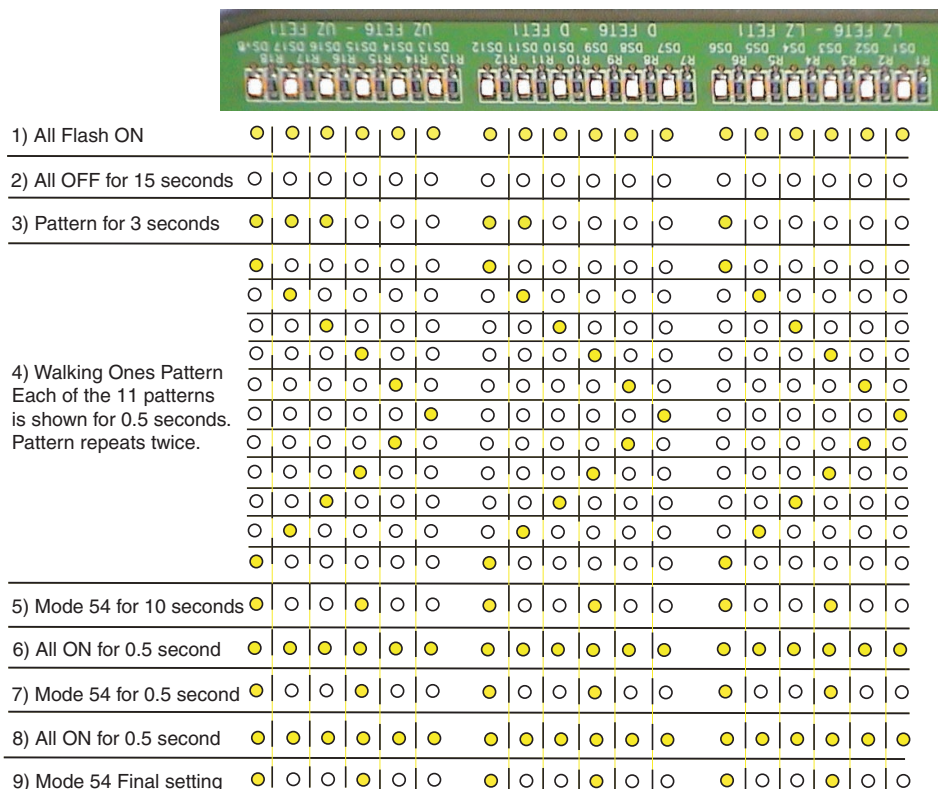
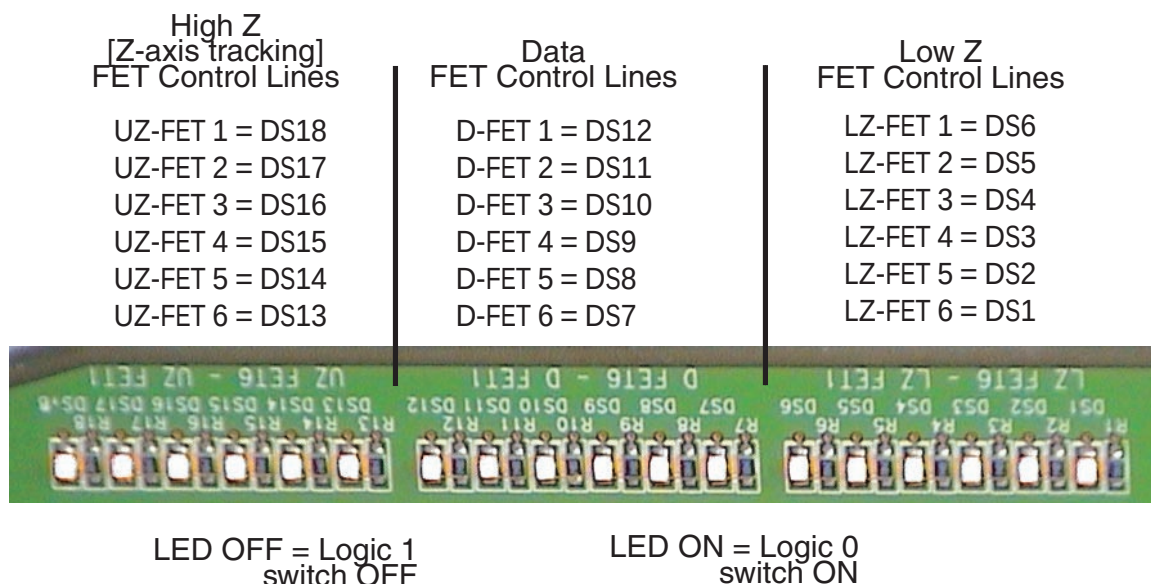


Figure 1-4 FET Control Interconnect

Section 4.0

DAS Backplane LEDs and Self-test



Section 5.0

DAS Serial Stream Data Transfer

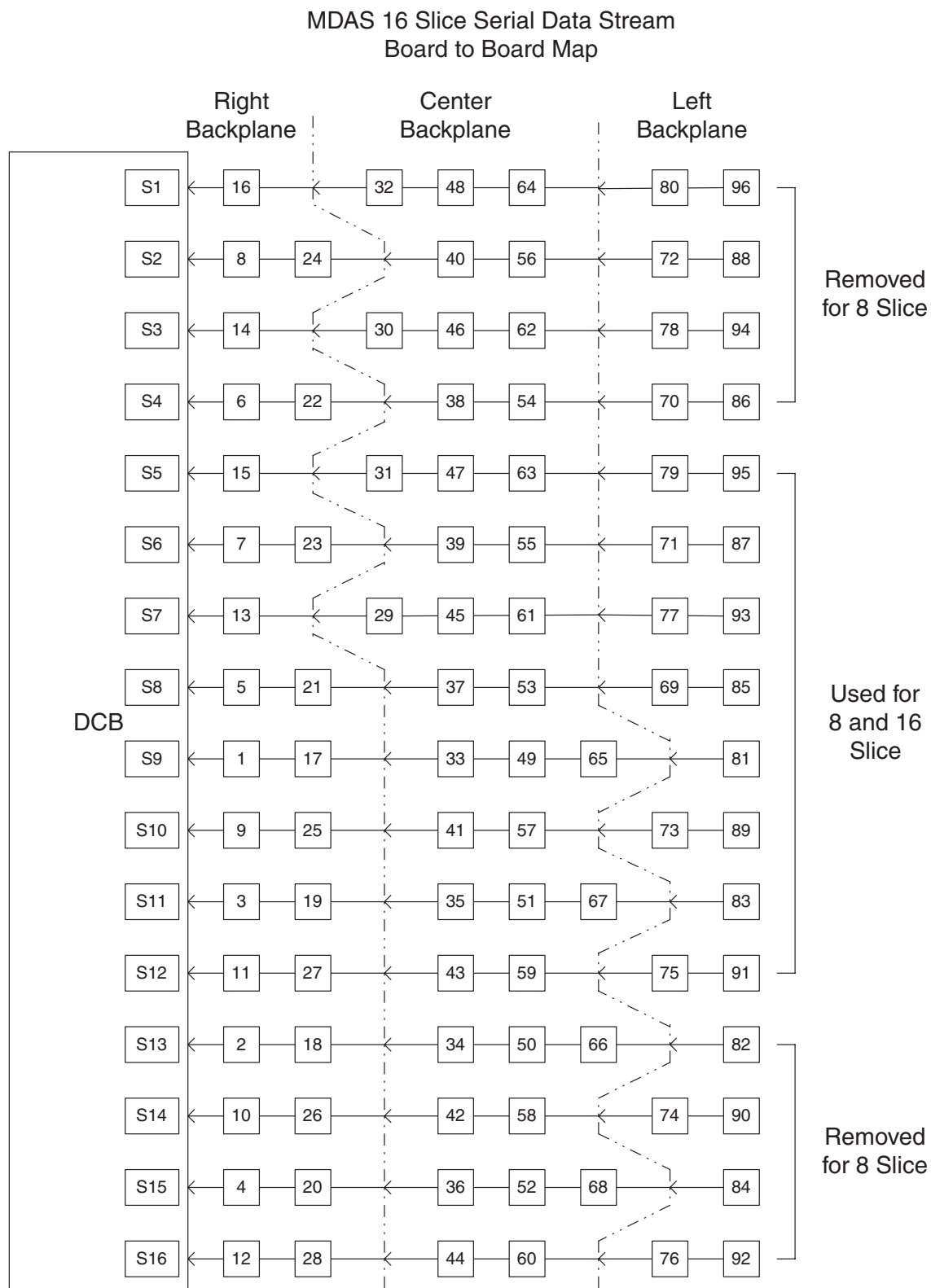


Figure 1-7 MDAS16 Serial Data Stream

Section 6.0

Detector Flex Pinout Diagram

Example: 8A16, 22 translates to -> Row 8, A side, Cell 16, Flex pin 22

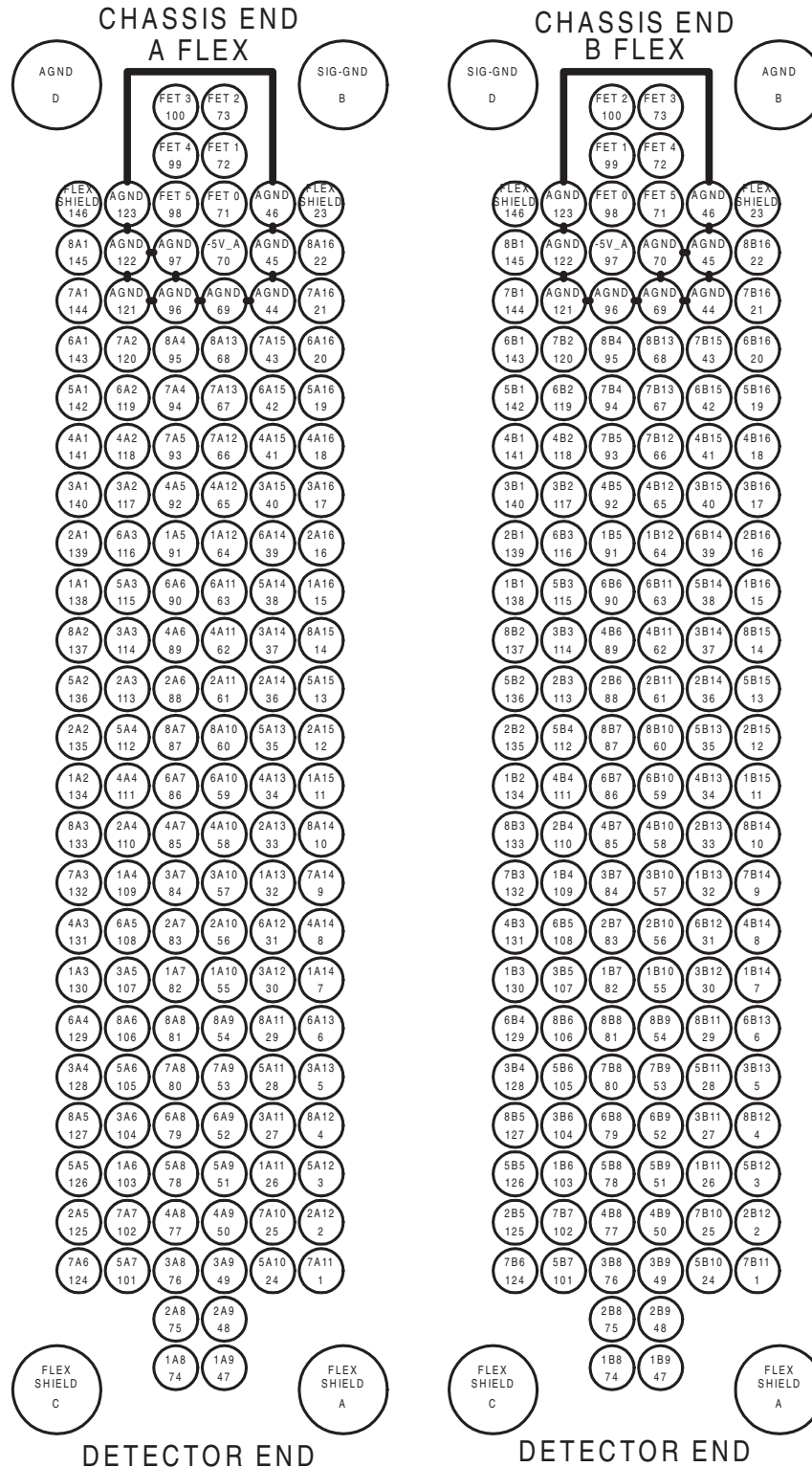


Figure 1-8 MDAS 16 Backplane Flex Pin Assignments

Section 7.0

DAS Power Supply Layout Diagram

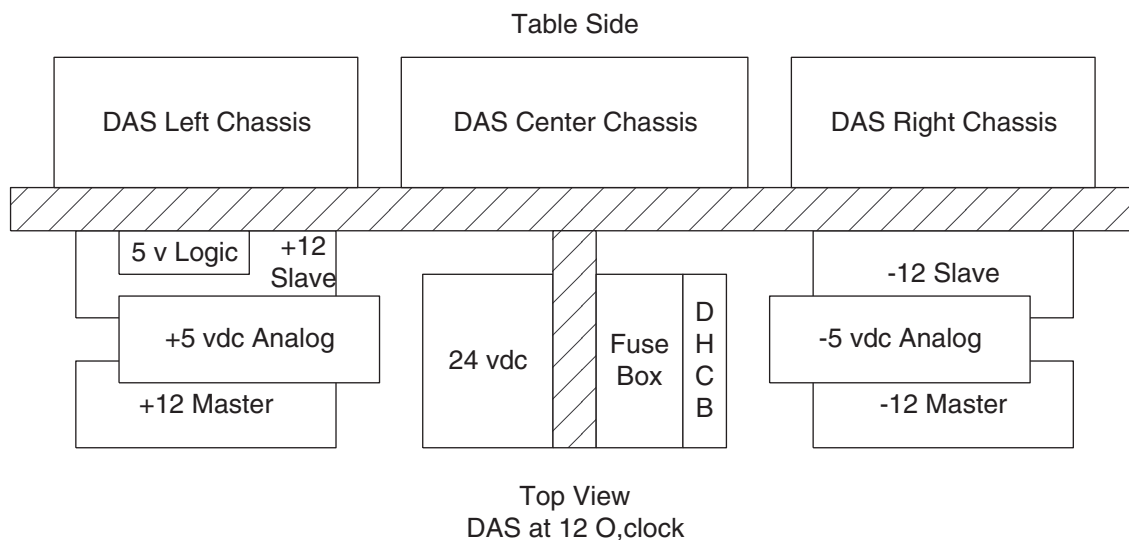


Figure 1-9 LightSpeed 16 Power Supply Layout

DAS Detector Architecture Map

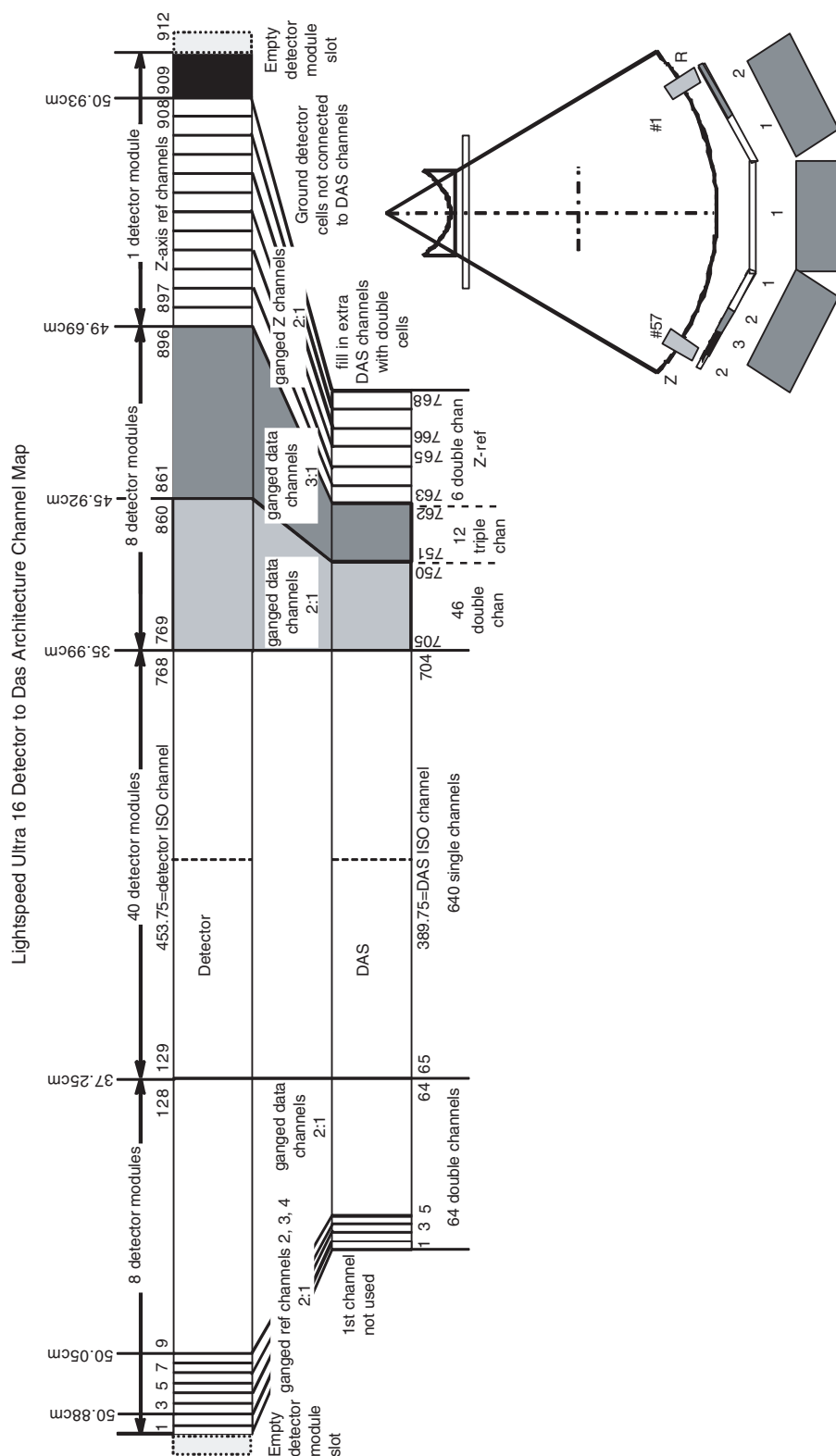


Figure 1-10 LightSpeed Ultra Detector to DAS Architecture Map

Section 9.0

DAS Detector Channel Map (Eye Chart)

The DAS-Detector graphical architecture drawing is intimidating to view and understand at first glance. However, it is easily mastered with a little explanation of its layout. The drawing is annotated with explanations.

- Each detector row will have the same type of description.
- Detector modules touch all Detector rows.
- Converter cards only touch two Detector rows unless they are the outer most cards. They then touch four Detector rows.
- The QIF tool is great for seeing the relationship between out of spec physical locations but it compresses all the detector rows into one dimension.
- The “eye chart” is helpful when you want to see how things vary across Detector rows (in the Z axis) or to help interpret a Scan Analysis plot without the QIF tool being up.

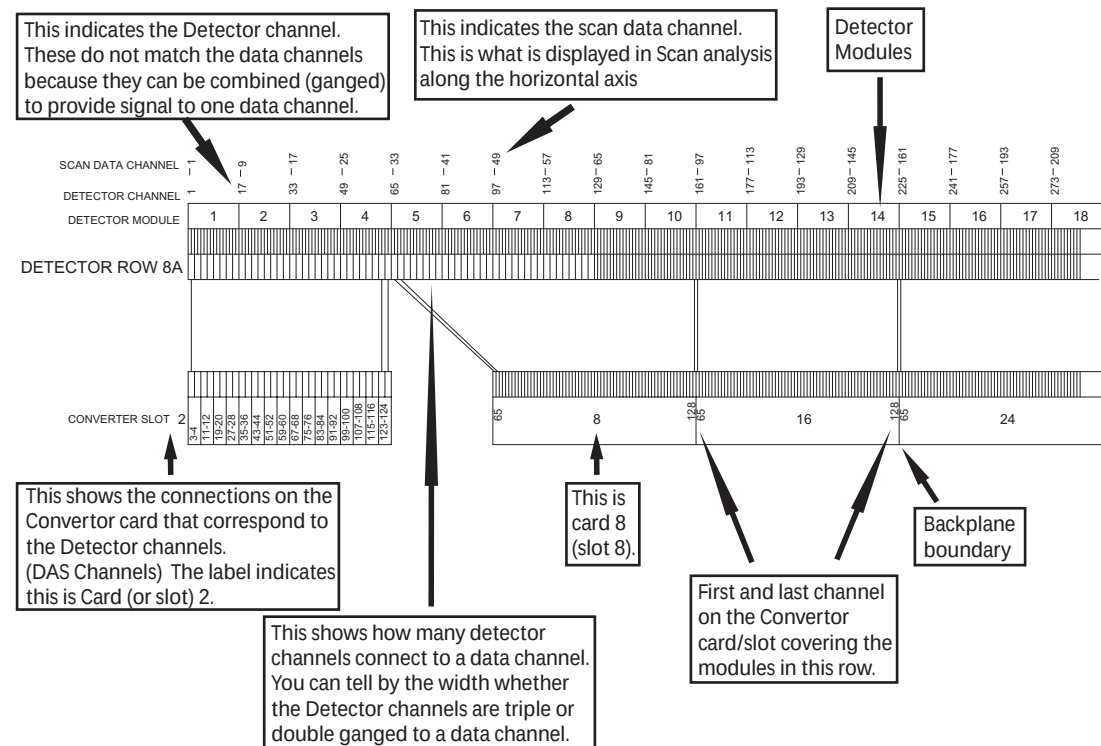
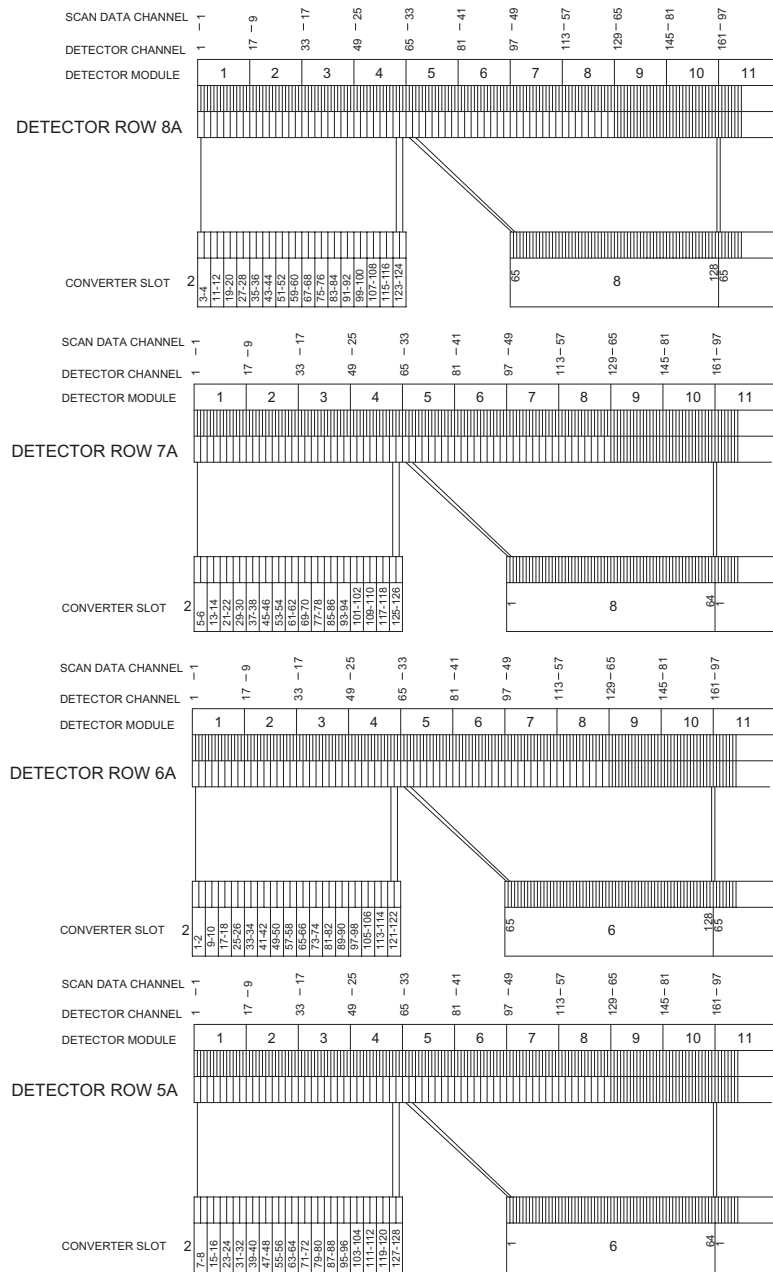


Figure 1-11 DAS Detector Channel Map Definitions



This shows card 2 covers 4 Detector rows on the end where there are ganged Detector channels.

Card 8 covers two rows and card 6 covers two rows for the same set of modules on the "A" side of the Detector.

This same pattern is carried out on the "B" side.

Figure 1-12 DAS Detector Channel Map Definitions, continued

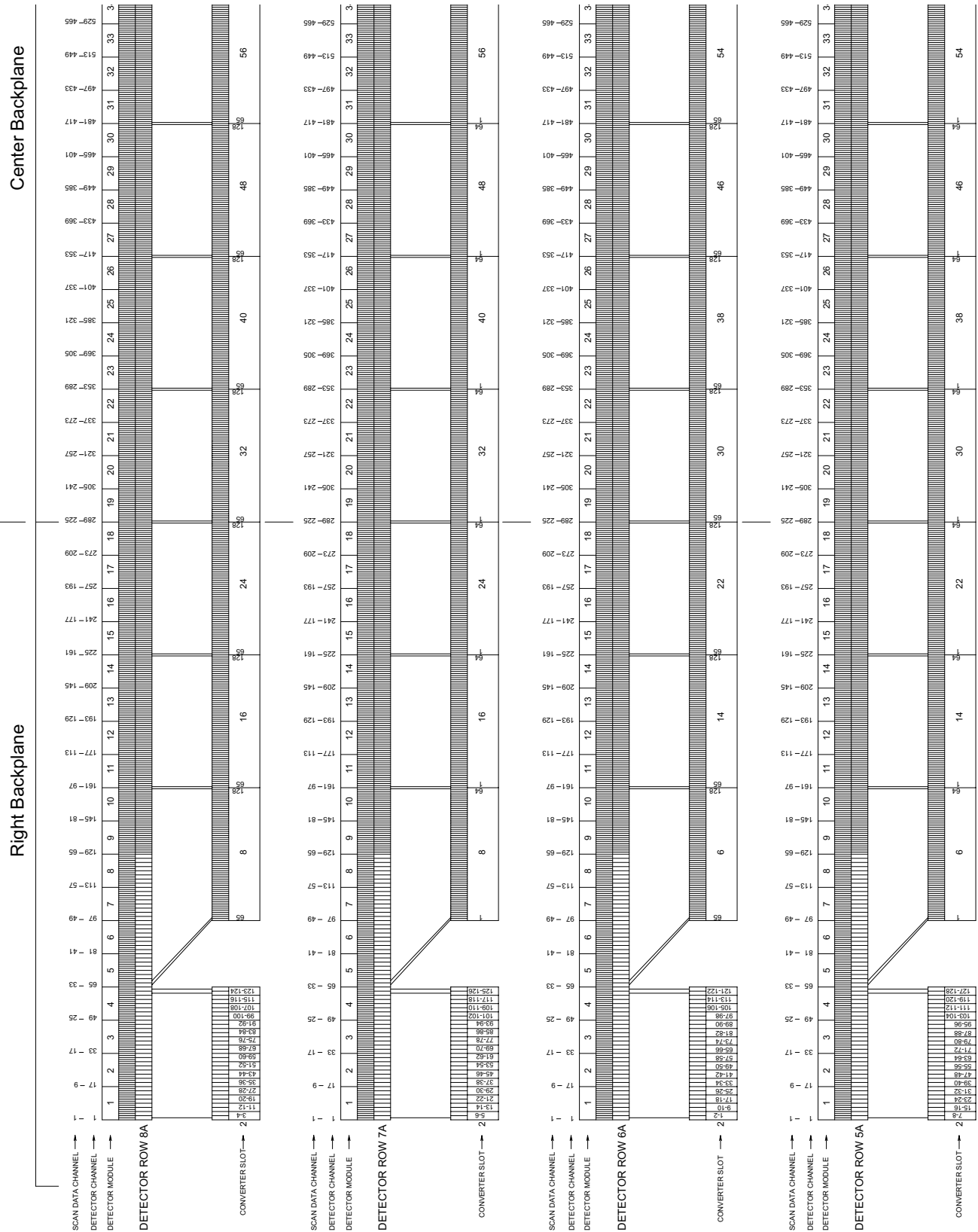
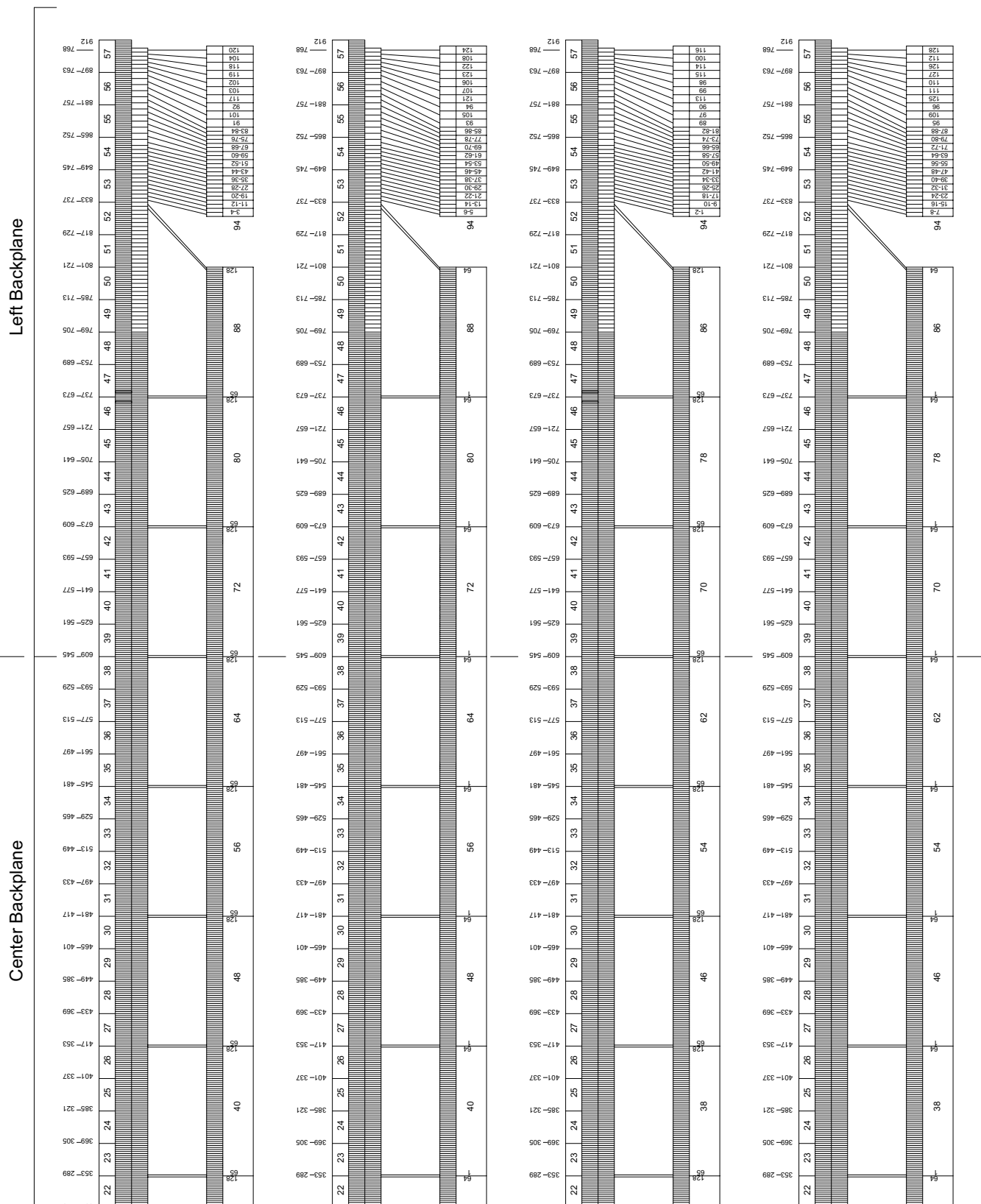


Figure 1-13 MDAS 16 Detector-to-DAS Map, Rows 8A, 7A, 6A & 5A



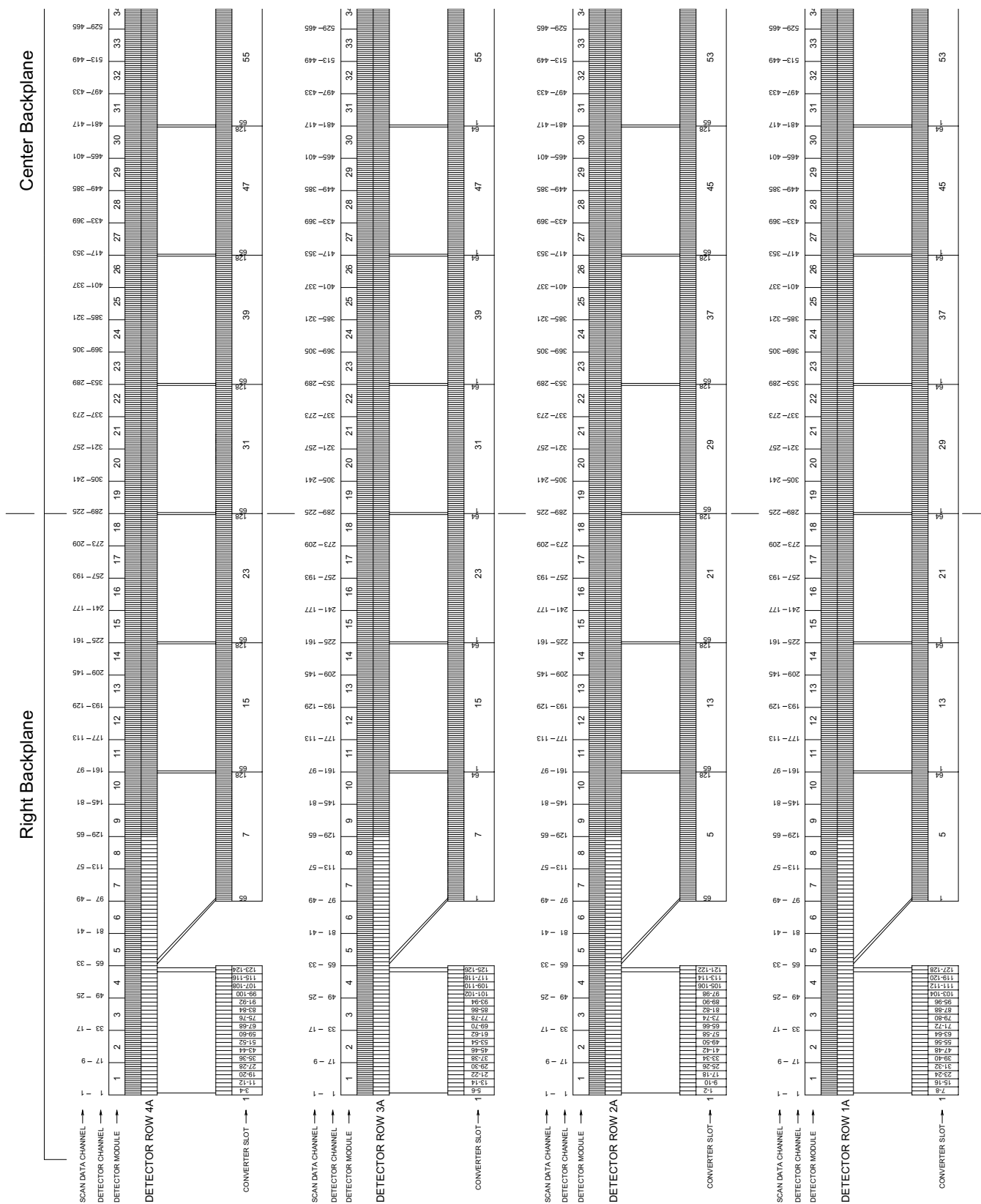


Figure 1-15 MDAS 16 Detector-to-DAS Map, Rows 4A, 3A, 2A & 1A

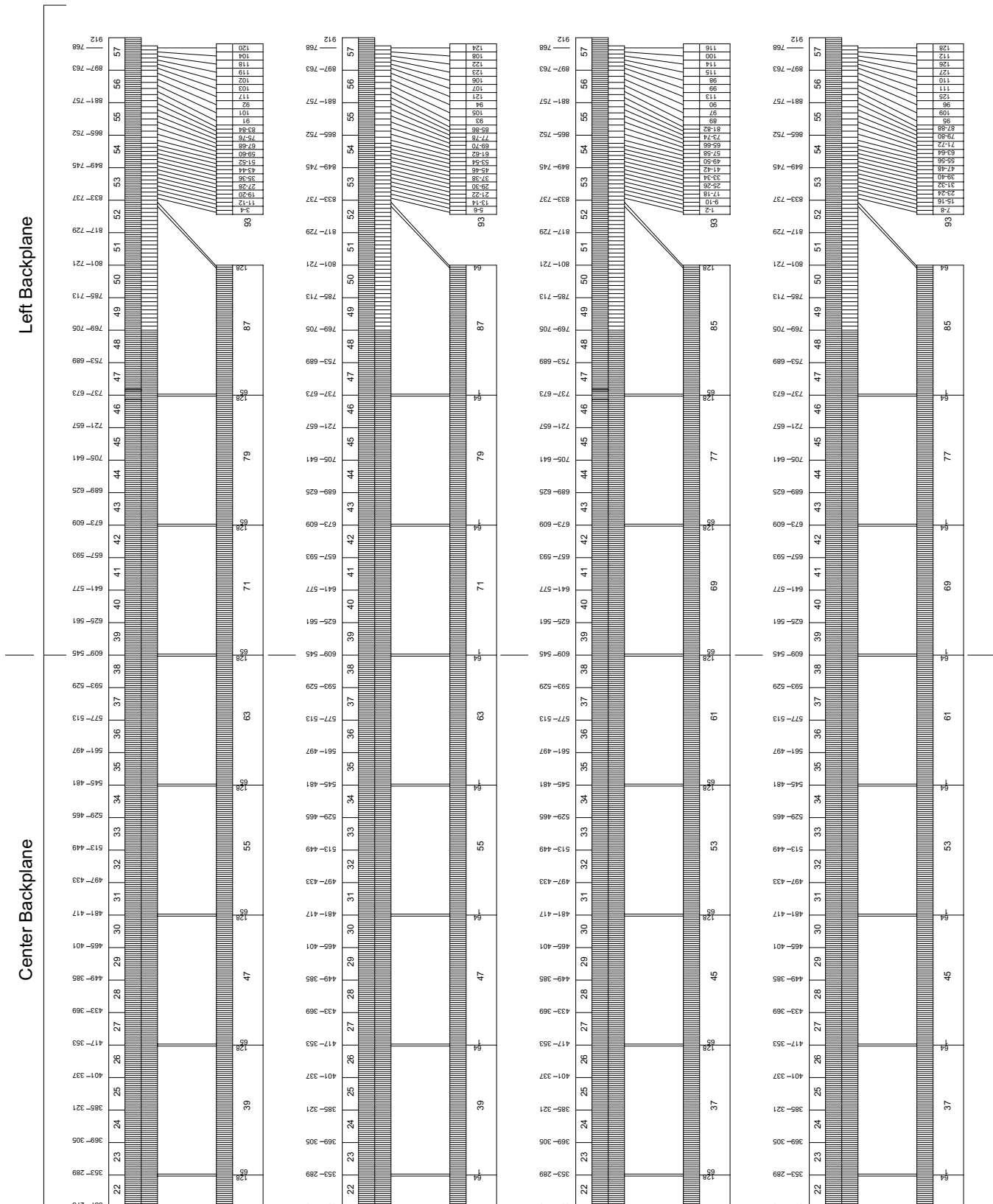


Figure 1-16 MDAS 16 Detector-to-DAS Map, Rows 4A, 3A, 2A & 1A, Continued

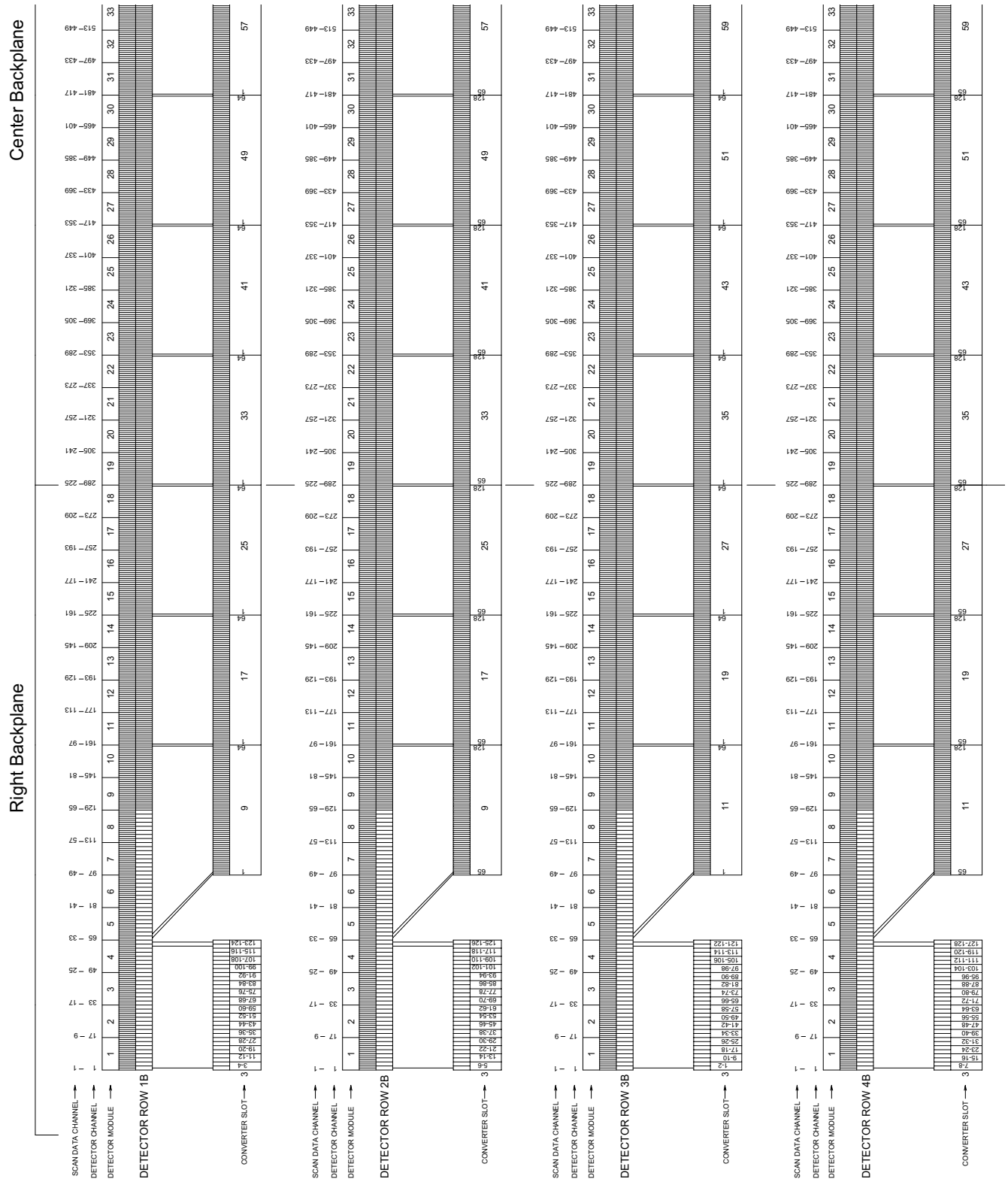


Figure 1-17 MDAS 16 Detector-to-DAS Map, Rows 1B, 2B, 3B & 4B

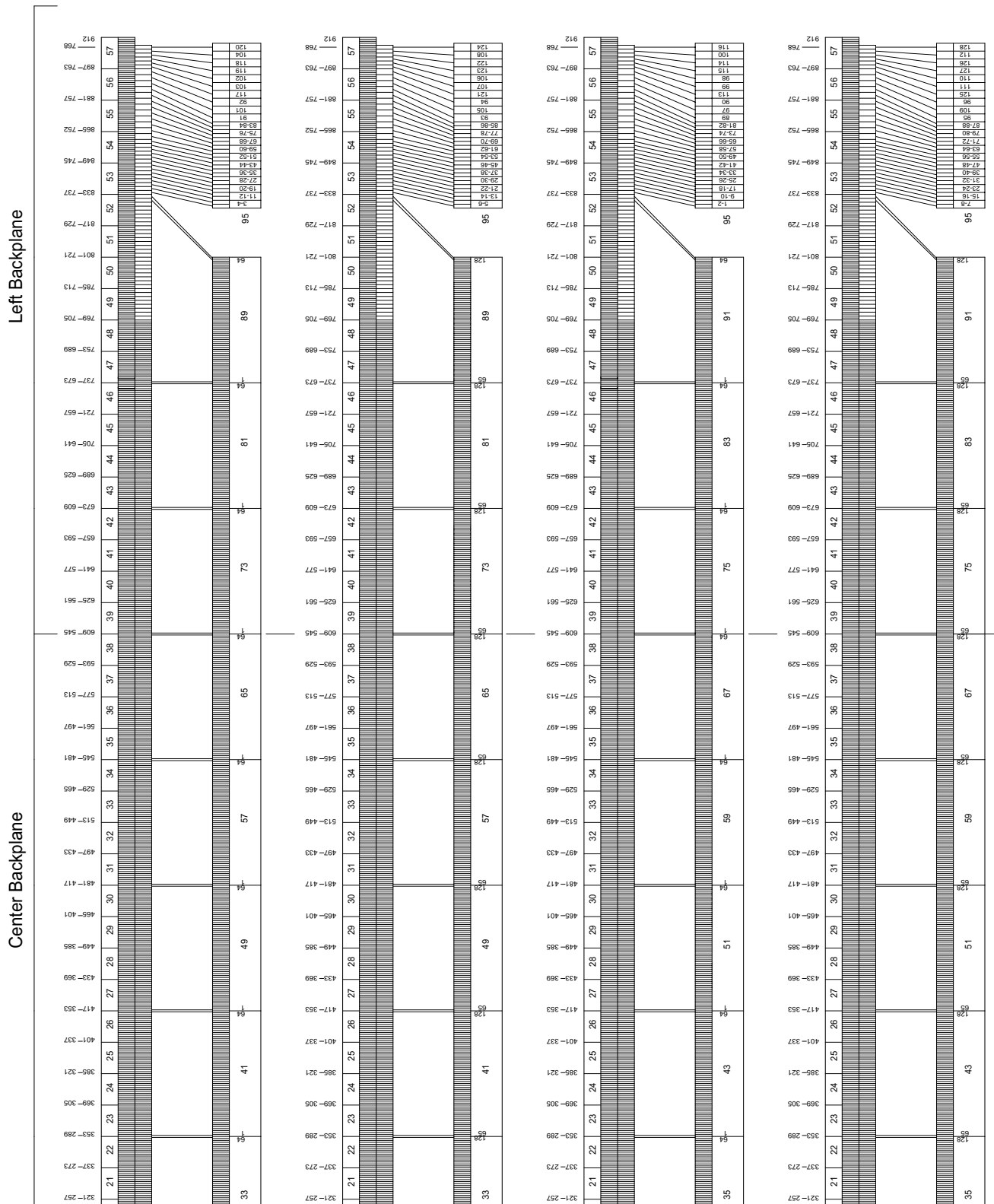


Figure 1-18 MDAS 16 Detector-to-DAS Map, Rows 1B, 2B, 3B & 4B Continued

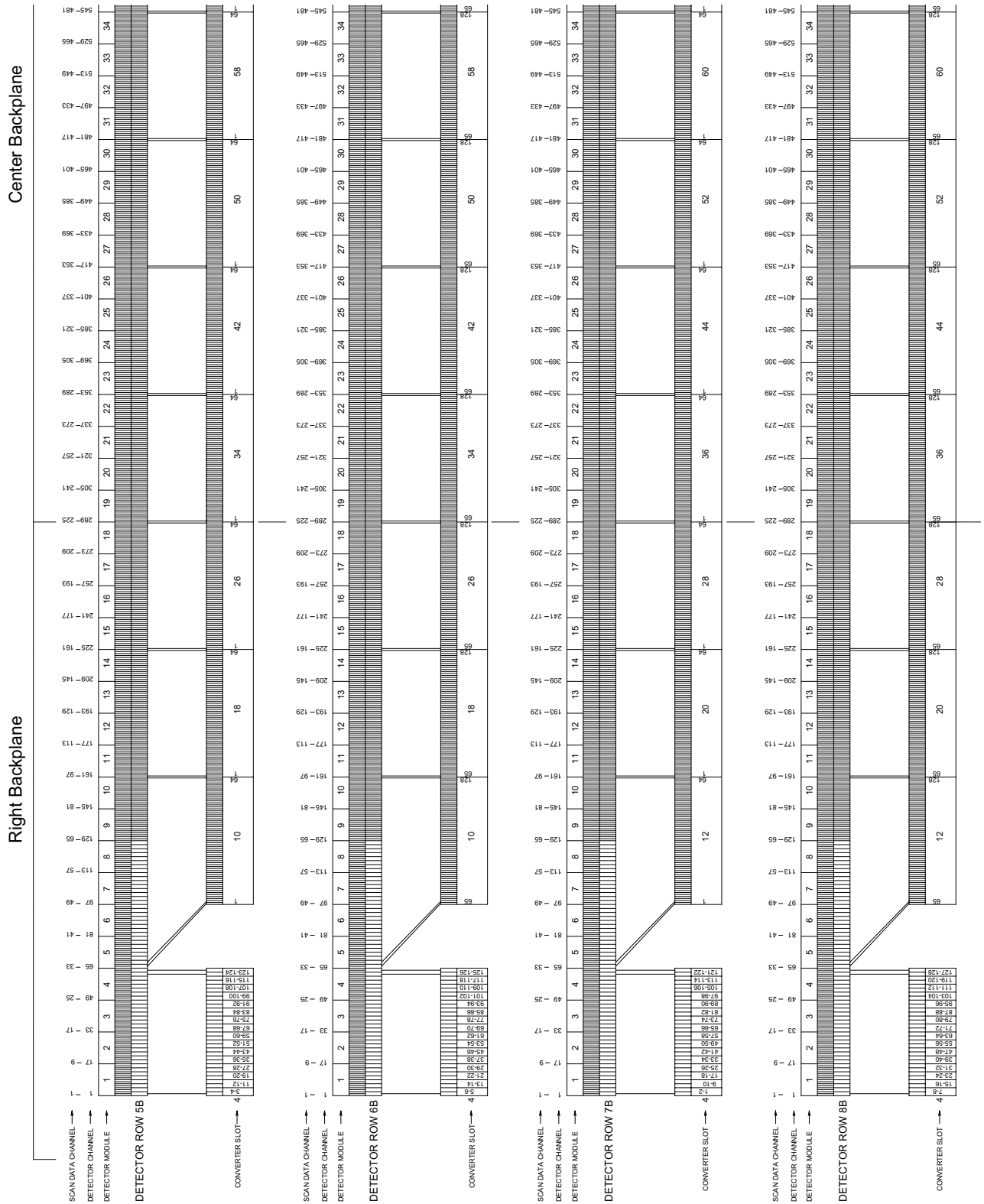


Figure 1-19 MDAS 16 Detector-to-DAS Map, Rows 5B, 6B, 7B & 8B

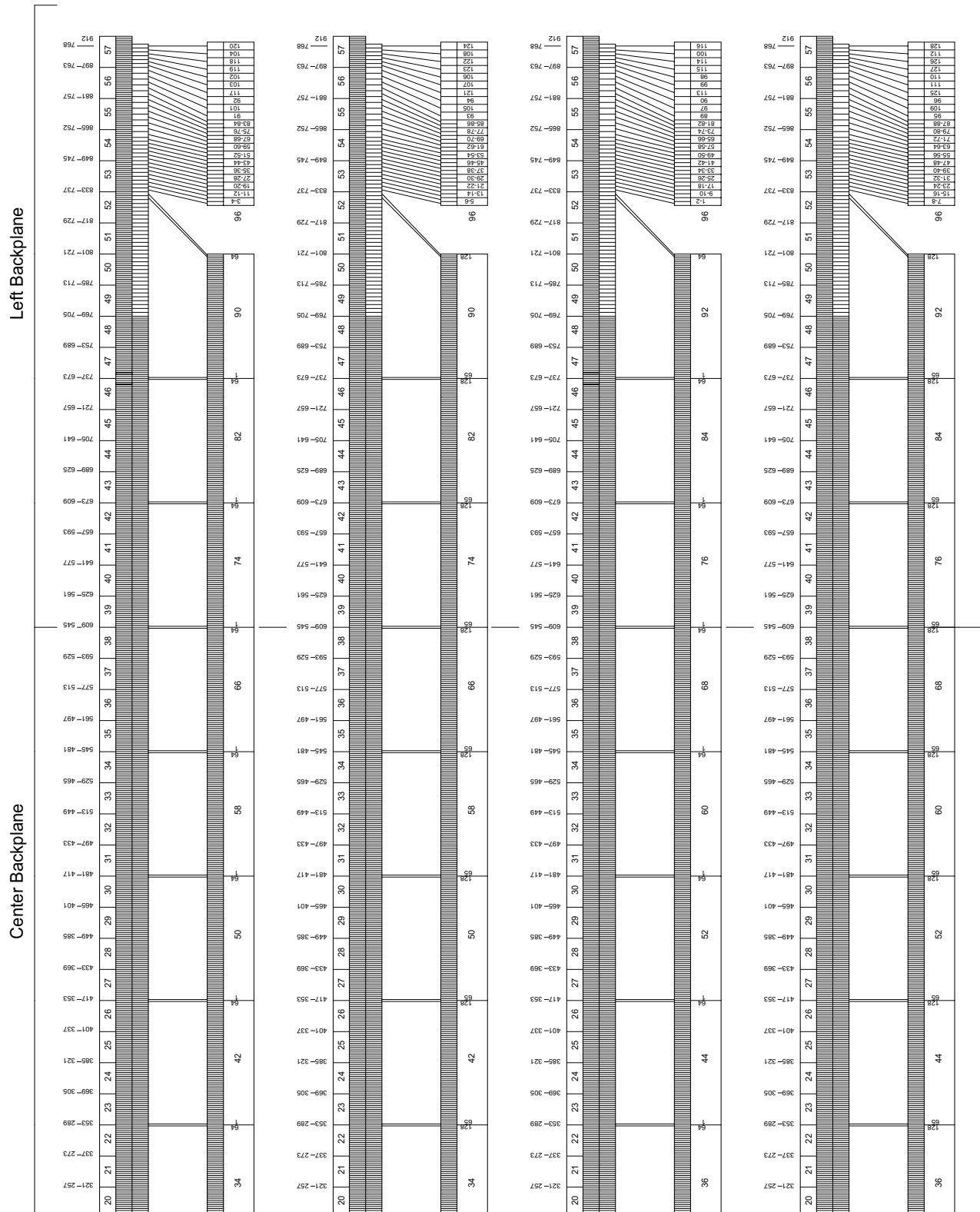


Figure 1-20 MDAS 16 Detector-to-DAS Map, Rows 5B, 6B, 7B & 8B Continued

Chapter 2

Troubleshooting Theory

Section 1.0

Image Artifact Definitions

1.1 Steak Artifact Theory

- Streaks are caused by a system channel or channels that deviate relative to neighboring channels for a single or very short number of views.
- Generally the result of Pop Noise events.

1.2 Rings Artifact Theory

- Rings are caused by a system channel or channels that deviate relative to neighboring channels as view data is collected.
- If deviation occurs from one to a very short number of views, then streaks are produced.
- If deviation occurs for the partial views, then partial rings or arcs are produced in Axial scanning.
- If deviation occurs for all views, then complete circles or rings are produced in Axial Scanning.
- For helical scans, arcs are generally full rings. These arcs can be visible across several concurrent images. This is due to the view weighting algorithm used in helical reconstruction. Analyze the middle image or most intense artifact.

1.3 Bands Artifact Theory

- Bands are caused by collection of adjacent channels that deviate relative to neighboring channels as view data is collected.
- If deviation occurs for the partial views, then partial bands or arcs are produced in Axial Scanning.
- If deviation occurs for all views, then complete circles or bands are produced in Axial Scanning.
- For helical scans, arcs are generally full bands. These arcs can be visible across several concurrent images. This is due to the view weighting algorithm used in helical reconstruction. Analyze the middle image or most intense artifact.

1.4 Z-Tracking Artifact Theory

- Unknown. System design is such that tracking failures are sensed by the design. In the event that the tracking failure is not sensed, artifacts can occur. However this is due to a hardware failure, generally actual FET mode is different from requested FET mode.
- No confirmed Z-Tracking Artifacts identified since introduction of the closed loop post patient beam collimation.

1.5 Detector Radiation Damage

- Banding artifacts that align precisely with detector module boundaries and are in consecutive images.
- Caused by excessive radiation exposure to the detector in a short period of time. Artifacts manifest themselves as follows:
 - Extremely high dose received by the detector.
(This is far beyond normal or even abnormal patient scanning techniques.)
 - Full fastcal performed.
 - 8 to 24 hours elapse without another full fastcal.
- By definition, this is a normal function of the Lumex material used in detector construction. The title is misleading. This is one reason fastcal is performed, to refresh cal vectors for Lumex gain drift. This type of artifact is very rare. Before condemning the detector, get Detector Engineering involved to evaluate the detector performance.

1.6 Center Smudge

Dark or light area of 169 pixels (13 x 13 box) near the center of scan FOV having poorly defined edges. The reference area around the smudge is measured by the ROI of a 41 x 41 ellipse and does not include the smudge area.

1.7 Center Artifact

A sharply defined small area (usually the center 4 pixels) having mean pixel values that differ more than a threshold value (see below) from the reference area. The average of a four pixel box that includes at least one of the four center pixels must be greater than the specified threshold value to be considered a failure.

1.8 Clump

Small light or dark areas at the center of the scan FOV. These must be 3 or more contiguous failing pixels within the center 9x9 pixels. All 3 of the pixels must be on the positive or negative side of the specification to be considered a failure.

1.9 Center Spot

Dark or light area near the center of the scan FOV having no defined edges and consisting of up to 25 pixels.

Section 2.0

Artifact Analysis Techniques

The following sections are based upon general troubleshooting concepts, software tool design, functionality and capability. With the design of today's systems, the volume of data that potentially contains useful analysis information is staggering. The information provided will help you to quickly identify the fault, isolate the fault condition and identify the faults root cause.

2.1 Basic Concepts

- 1.) Attempt to analyze the original patient image and scan data if possible.
- 2.) Ask the customer to duplicate the artifact using a phantom.
 - This step is overlooked by many Engineers.
 - Protocol choices and system setup knowledge is extremely important to be successful.
- 3.) Use both Large and Water Phantoms when reproducing an artifact.
 - Material density can play a factor in artifact generation.
 - Remember to scan and display the large field of view.
- 4.) Write everything down.

The biggest mistake is not documenting what you KNOW, and what you DID.

2.2 Basic Information

Before you start to analyze, gather the following information. [See section # 3.1 on page 34](#) for definitions and usage of Scan Analysis tools.

- 1.) Exam, Series, Images that artifact is present.
- 2.) Slice profile, 16 x 0.625, 8 x 1.25, etc.
- 3.) KV and MA
- 4.) Scan speed, 0.5, 1.0 seconds, etc.
- 5.) Cradle speed (Pitch)
- 6.) Head First or Feet first.
- 7.) Cradle direction, in or out of the gantry.
- 8.) SFOV (Scan field of view) and DFOV (Display field of view).

2.3 Defining the Problem

This section is a basic set of questions to help identify WHAT you want to fix. The possible failure modes are not ranked or all inclusive. There are no absolutes, just logical progressions. These questions should help to guide you in how to approach the failure.

2.3.1 Artifacts

- Does the artifact occur in ALL images?

If yes then problem is most likely not within the DAS.

 - Possible recon, power supply, DCB, DIP, or detector.
 - DASTOOLS-> DC Noise and Offsets might be useful.
 - Scan Analysis-> Aux Channels, Z-Axis Channels might be useful.
- Does the artifact occur in one or more adjacent images?

If yes then problem is most likely not within the DAS.

- Possible recon, power supply, detector, beam obstruction.
- DASTOOLS-> Interconnect/Xray Ver might be useful.
- Scan Analysis-> Aux Channels, Z-Axis Channels, Cal Vectors might be useful.
- DDC -> Air scans for Scan Analysis.

- Does the artifact occur once every ? images where ? is (? x 1.25), (? x 0.625), (? x Zmm).

If yes then the problem is most likely in the DAS.

- Possible convertor card, DDIF, detector, static charge or debris, beam obstruction.
- Scan Analysis-> VVC, MSD, Cal Vectors most useful.
- DDC -> Air scans for Scan Analysis MSD plot, DD file ratio, film on collimator.
- DASTOOLS-> DC Noise and Offsets, Interconnect/Xray Ver might be useful.

- Does artifact change from exam to exam or image to image?

If yes then the problem is most likely not in the DAS.

- Possible power supply, static charge, debris, beam obstruction, intermittent detector module, EMI.
- Scan Analysis-> VVC with FFToffset corrected, MSD, Aux Channels, Z-Axis Channels most useful.
- DDC -> Air scans for Scan Analysis MSD plot, DD File ratio.
- DASTOOLS-> DC Noise and Offsets, Interconnect/XRay Ver, Pop Noise & Microphonics might be useful.

- Does artifact minimize or disappear after Fastcal or Detailed Cal and then reappear?

- Save current cal set to new State MOD.
- Reload known good cal files from State MOD.
- Reproduce Artifact and evaluate.
- Never troubleshoot artifacts with corrupted cals. It's hard enough without fighting software as well.

2.3.2 Specification or Limit Errors

Below are two of the most common faults encountered. This is not a complete list.

2.3.2.1 Beam Tracking Related Errors

These errors are reported by the DCB. This is generally not a Collimator Hardware issue. It is most likely one of the following.

- Tube alignments are the most frequent cause.
- Beam obstruction.
- Faulty convertor card.
- Faulty DCB.

2.3.2.2 Specific Aircal Processing Error

Mon Sep 23 07:46:49 2002

Error: 230017082

Host: ctradio

Process: accs

File: aircal_processing.c

Line: 898

Channel in non-bowtie aircan out of range - row 7, channel 1, value = 3.5582

This error is presented from the point of view of a software programmer. Below is an explanation of how to decode this error. Please note that the processing of data stops with the first failure. When this specific problem is corrected, you may have the same failure in a different row or channel. **Read the following carefully or you will become hopelessly lost and confused.**

- Detector row number starts with Zero (0) and ends with number of data rows collected. For LightSpeed 16, this means row 0 through 15. (0 – 7 for 8-slice Ultra, 0 – 3 for 4-slice QX/I.)
 - 0 = 8A, 15 = 8B
 - 1 = 7A, 14 = 7B
 - 2 = 6A, 13 = 6B
 - 3 = 5A, 12 = 5B
 - 4 = 4A, 11 = 4B
 - 5 = 3A, 10 = 3B
 - 6 = 2A, 9 = 2B
 - 7 = 1A, 8 = 1B
- Channel number is the first “data” detector channel processed, again from a software programmers point of view. The programmer is attempting to report the Detector Channel. This means you must use the Detector-to-DAS Maps, beginning on [page 19](#), to decode the DAS channel correctly. This is because of channel ganging and unused channels. **Failure to use these maps will lead you on a wild goose chase.**

Physical Detector Channel = Reported Channel + 8
- The value is unimportant since it is out of spec and needs to be corrected.

[Table 2-1](#) is a translation table example.

REPORTED ERROR		PHYSICAL TRANSLATION		
Row	Channel	Row	DET. CHANNEL	DAS CHANNEL
0	1	8A	9/10	5
1	119	7A	127/128	64
2	121	6A	129	64
3	445	5A	453	389
4	760	4A	768	704
5	761	3A	769/770	705
6	851	2A	859/860	750
7	853	1A	861/862/863	751
8	886	1B	894/895/896	762
9	889	2B	897/898	763
15	899	8B	907/908	768

Table 2-1 Example Translation Table

Once you have performed this translation you can verify your work by using the DAS/DETECTOR ARCHITECTURE MAP found in DASTOOLS or Scan Analysis MSD Plot window Tool Pull-down.



NOTICE

The DAS/DETECTOR ARCHITECTURE MAP accepts DAS channels only. You cannot get the correct DAS channel using this software tool only.

2.3.3 Ring Radius Calculation

The following formula is an approximation only. The accuracy goes down the further away from ISO Center the ring is. This should be used to ballpark where the artifact might be.

1 mm = 1.75 DAS channels from ISO.

ISO = 389.75

If the ring radius is 35 mm from ISO then the possible channels are:

$35 \times 1.75 = 61.25$

Lower Channel = $389 - 61 = 328$

Upper Channel = $389 + 61 = 450$

Remember, if you are off by 1 channel, you can miss the mark by 4 convertor cards. Don't waste time and money, use the tools available to you.

2.4 Reproducing the Artifact

For the LightSpeed Ultra 16 Slice you must remember that the detector has 24 rows but only 16 images can be produced in one revolution. See Figure 1-2 on page 9. This table illustrates the various FET modes used in patient scanning. Lets state the obvious. The center 16 detector rows are 0.625mm cells. Eight rows for the "A" side and eight rows for the "B" side. The four outside rows are 1.25mm cells. Four on the "A" side and four on the "B" side. Observe the difference in column width between A8 (0.625mm) and A9 (1.25mm).

What is not obvious is that you can effectively isolate many artifacts with 3 basic data collection protocols. 16 x 0.625mm, 16 x 1.25mm, 8 x 2.50mm. These three application FET mode configurations give you full detector coverage and simple macro row generation. This is very useful when the artifact appears once every (? x Zmm).

Note, helical scans can produce artifacts on 3 or more adjacent images. Remember, partial rings and bands in helicals are full ring or bands in axials. Always use axial scanning to reproduce artifacts.

2.4.1 Using New Patient

- Always start with the large SFOV and DFOV, both 50cm.
 - You need to see the entire image plane when troubleshooting.
 - There are three basic phantoms, large, water, and air.
 - * Scanning a centered large phantom first is the simplest technique.
 - * Scanning a centered QA phantom against a large FOV can reproduce failures near 12.5cm ring radius. (Expose through both water and line pairs).
 - * Scanning an off center QA phantom against a large FOV can reproduce subtle rings seen in lung studies. (Expose through both water and line pairs).
 - * Scan with the gantry tilted to further stress the imaging subsystem.
 - Adjust the window and level to see the air around the phantoms.
- Scan with both cold and hot tubes.

Artifacts that appear, disappear, or move with changes in temperature are most likely due to a beam obstruction. Investigate tube port and collimator for beam obstructions.
- Always scan supine, head first, table moving out of the gantry.

Consistent methodology reduces the chances of simple mistakes.

This method works very well with Scan Analysis VVC plots. Make sure you use offset correction and convolved data.

2.4.2 Using DDC

If you cannot reproduce the artifact using New Patient then DDC might help. DDC is useful for the following:

- Progressive system noise control during data collection.
 - Static, xray off, rotor off
 - Static, xray off, rotor on
 - Rotating, xray off, rotor off
 - Rotating, xray off, rotor on
 - Static, xray on, small or large filament, FET override off
 - Rotating, xray on, small or large filament, FET override off
 - Static, xray on, small or large filament, FET override on
 - Rotating, xray on, small or large filament, FET override on
- Detector profile data collection
 - Static, xray on, small or large filament, FET override off
 - Use Scan Analysis, MSD Plot, offset corrected (user preference) to evaluate.
- Data Collection for DD File Ratio analysis. Good for non linearity issues.
 - Scan 1, low technique stationary X-Ray On scan.
 - Scan 2, same as scan 1 except increased mA.
 - Use Scan Analysis, Create MSD DD File.

Section 3.0 Scan Analysis

3.1 Scan Header

The following is a portion of the scan header file. Comments are added for clarification as needed.
[Comments look like this].

```
EXAM_LEVEL_INFO: value = { [Section start]
  SUITE_ID: value = "CT48"
  PRODUCT_ID: value = "LightSpeed16"
  HOSPITAL_NAME: value = "G.E. Medical Systems"
  EXAM_NUMBER: value = 50131
SERIES_LEVEL_INFO: value = { [Section start]
  SERIES_NUMBER: value = 1
  SERIES_TYPE: value = SERVICE_AXIAL_SCAN_SERIES [Define Axial or Helical Series]
  PATIENT_POSITION: value = PATIENT_POS_SUPINE
  PATIENT_ENTRY: value = PATIENT_ENTRY_HEAD_FIRST
SCAN_LEVEL_INFO: value = { [Section start]
  SCAN_TIME_SEC: value = 1.0000
  OVERRIDES: value = { [Section start]
    DATACHANNELCONTROL: value = FET_4x1_250 [This is the FET Mode]
    ZCHANNELCONTROL: value = FET_4x1_250_Z_MODE [This is the Z-FET Mode]
    ~~~~~
  Z_AXIS_TRACKING: value = TRACKING_MODE_OFF [Section start]
  KVOLT: value = 80
  MAMP: value = 0
  SPOT_SIZE: value = FOCAL_SPOT_SMALL
  DET_MACRO_WIDTH_ISO: value = MACRO_ROW_WIDTH_1_25MM [Should be the same as the ZCHANNELCONTROL from above.]
  NUMBER_DETECTOR_MACRO_ROWS: value = MACRO_ROWS_4 [Should be the same as the ZCHANNELCONTROL from above.]
  GANTRY_FILTER: value = FILTER_BODY
  TRIGGER_FREQUENCY_HZ: value = 984.0000
  TABLE_START_MM: value = 0.0000
  TABLE_END_MM: value = 0.0000
  TABLE_SPEED_MM_SEC: value = 0.0000
  SCAN_PITCH_RATIO: value = "None"
  GANTRY_PERIOD_SEC_REV: value = 1.0000
  GANTRY_TILT_DEG: value = 0.0000
  SCAN_FOV_TYPE: value = SFOV_LARGE
  SCAN_FOV_MM: value = 500.0000 [Value in millimeters]
  EQUIPMENT_INFO: value = { [Section start]
    ~~~~~Lots of output~~~~~
  SIZEOF_VIEW: value = 6248
```

TABLE_DIRECTION: value = TABLE_MOVE_INTO_GANTRY

~~~~~

RECON\_REQUEST\_VALID: value = YES **[Section start]**

IMAGE\_THICKNESS\_MM: value = 2.5000

IMAGE\_LOCATION\_INCREMENT\_MM: value = 0.0000

DFOV\_MM: value = 250.0000

TARGCEN\_R: value = 0.0000

TARGCEN\_A: value = 0.0000

RECON\_REQUEST: value = { **[Section start]**

FIRST\_SUPER\_VIEW\_FOR\_IMAGE: value = 9

NUMBER\_OF\_SUPER\_VIEWS\_FOR\_IMAGE: value = 984

FIRST\_VIEW\_FOR\_ROW: value = {

ROW0: value = 9

ROW1: value = 9

ROW2: value = 9

ROW3: value = 9

}

NUMBER\_OF\_VIEWS\_FOR\_ROW: value = {

ROW0: value = 984

ROW1: value = 984

ROW2: value = 984

ROW3: value = 984

VIEW\_COMPRESSION\_FACTOR: value = 1

DFOV\_MM: value = 250.0000

TARGCEN\_R: value = 0.0000

TARGCEN\_A: value = 0.0000

SIGMA\_B\_NUM\_ROWS\_USED\_PER\_IMAGE: value = 1

SIGMA\_B\_INDEX\_OF\_FIRST\_ROW\_USED: value = 0

## 3.2 Cal Vectors

In many cases the sin and cosine vectors will show intermittent artifacts. You need the original scan data which contains the artifact. Standard Deviations plot generally will show abnormal spikes better than the Means plot.

Reference the System Service Manual for normal Cal Vector plot examples.

## 3.3 Auxiliary Channels

### SCAN FILE AUXILIARY CHANNELS

|                             |                                                           |
|-----------------------------|-----------------------------------------------------------|
| Detector Heater Temperature | Reports only one zone of the detector. minimal usefulness |
| MA                          | Reports mA values during XRay on scans                    |
| KV                          | Reports KV values during XRay on scans                    |
| Positive 12 Volt            | Plots +12 vdc analog power supply                         |
| Negative 12 Volts           | Plots -12 vdc analog power supply                         |

## SCAN FILE AUXILIARY CHANNELS

|                                |                                                          |
|--------------------------------|----------------------------------------------------------|
| Positive 5 Volt Analog         | Plots +5 vdc analog power supply                         |
| Negative 5 Volt Analog         | Plots -5 vdc analog power supply                         |
| Positive 5 Volt Test Reference | Presents invalid data. Ignore this.                      |
| Negative 5 Volt Reference      | Presents invalid data. Ignore this.                      |
| 5 Volt Dc Power Supply         | Plots 5 vdc logic power sensed at the DCB Circuit board. |
| Positive 5 Volt Converter      | Presents invalid data. Ignore this.                      |
| Digital Ground                 | Data not valid. Ignore this.                             |
| Analog Ground 11               | Data not valid. Ignore this.                             |
| Analog Ground 12               | Data not valid. Ignore this.                             |
| Analog Ground 15               | Data not valid. Ignore this.                             |

### 3.4 Create MSD DD File

This function will create Diagnostic Data Files of any valid scan data file.

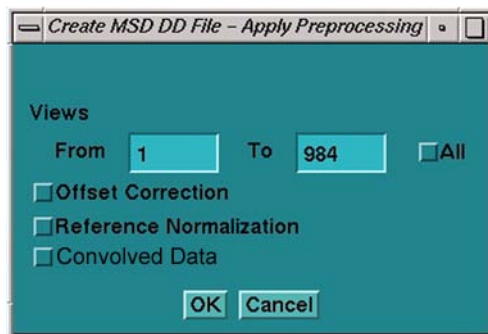
Refer to [2.4.2 - Using DDC](#) for suggested troubleshooting process.

Click on **UPDATE** if desired files are not present.

- Click on the desired exam/series/scan.
- Click on **CREATE MSD DD FILE**.  
Default all preprocessing.
- Select a second scan and repeat.

The above steps support DDC scans for DD File Ratio Analysis. The same analysis can be performed on New Patient data. The preprocessing options seen in [Figure 2-1](#) provide additional choices as follows:

- Offset Correction->Subtract offset data from DD File data.
- Reference Normalization->Normalizes XRay reference channels 2, 3, and 4. These channels are used to monitor mA or blocked channel occurrences.
- Convolved Data->Basically compares scan data against calibration files. Helps to bring out subtle changes not visible in raw data.



**Figure 2-1 Create MSD DD File Preprocessing Options**

**Note:** Make sure you clean up the DD files you create. The path is shown in the pop-up windows. More than 100 files can cause indexing problems.

### 3.5 Plot VVC (Views Versus Channels)

VVC is the most versatile and complex tool available. It is also difficult to identify many artifacts using VVC.

General tips;

- New Patient generated scan data.  
Use offset corrected and convolved data. This will help bring out issues not visible in raw data.
- DDC generated scan data.  
Use offset corrected or no corrections. Convolved is not valid since no Calibration files are associated with the raw data.
- Both scan data types.  
Use offset corrected and FFT to view Frequency versus Channel Plots. This can be useful in identifying strange noise or random artifacts.  
Remember FFT is very subjective. It is normal to see some white lines from left to right (Freq). It is the intensity that might indicate a problem. See [Figure 2-9 on page 48](#) and [Figure 2-10 on page 49](#) for examples of abnormal data.

### 3.6 DD Tab (Diagnostic Data)

DD File analysis cannot be used on scan data directly. The scan data must first be converted into a usable format. There are 6 basic functions, four basic math operations, Channel to Channel (Ch2Ch), and a Save to ASCII feature. The four basic math functions allow you to add subtract multiply and divide common file types. Save to ASCII creates a txt file format which can be exported in Excel for further manipulation. A very useful operation is to divide one file into another also called ratio of two files. Ch2Ch currently has little value for Service Troubleshooting, it is basically used for developmental research and data. Note, a variety of errors can be generated if the files are not of the same type or size.

To perform DD File Ratios do the following;

Refer to [2.4.2 - Using DDC](#) for suggested troubleshooting process.

Select Scan Tab from main screen, See [Figure 2-2 on page 38](#).

- Click on **UPDATE** if desired files are not present.
  - Click on the desired exam/series/scan.
  - Click on **CREATE MSD DD FILE**.  
Default all corrections.
  - Select a second scan and repeat.
- Select the DD tab at the Top of the Scan Analysis GUI.
  - Select Scan 1 of the msd DD files and click the File 1 button.
  - Select Scan 2 of the msd DD file and click the File 2 button.
  - Select the “/” or “Ratio” button.
  - A plot of the “Ratio” data should now be available for evaluation.
- Both “Plot” and “DD File” can be selected at the same time. This will activate the “Output Filename” field for descriptive naming as well as generating a plot

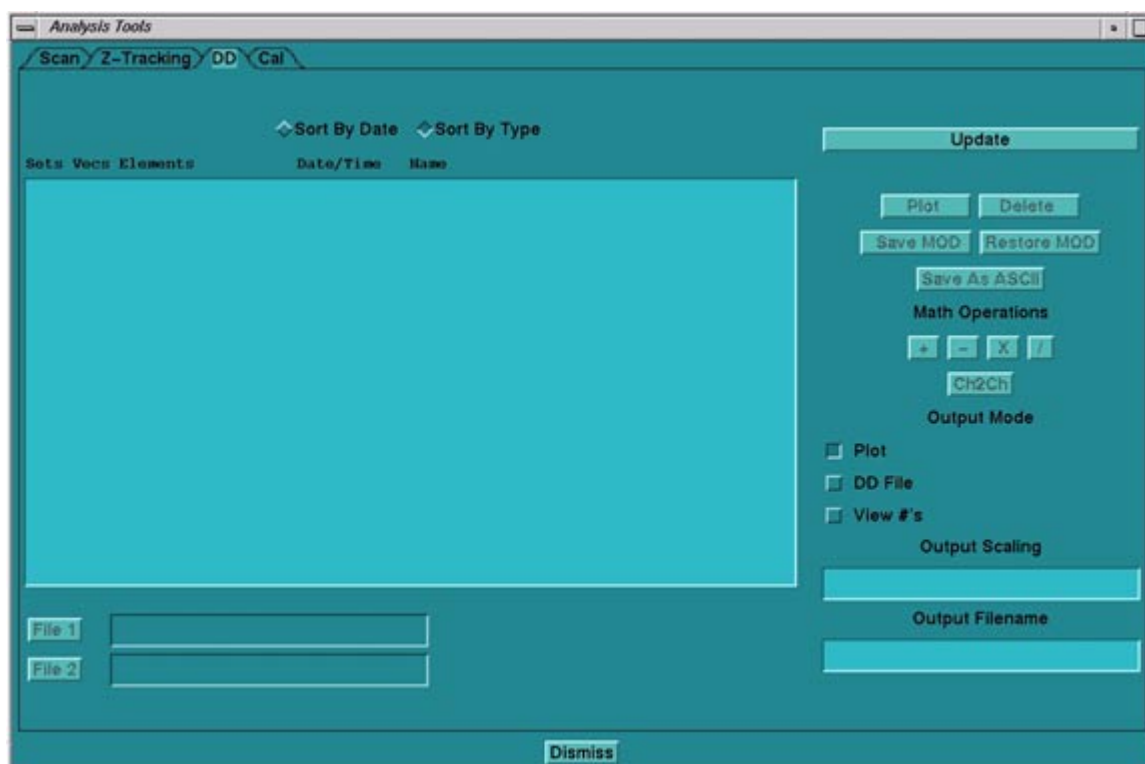


Figure 2-2 DD File Analysis Screen

## Section 4.0

# DDC Diagnostic Data Collection

DDC allows the user almost complete control over data collection. This is used heavily during product development. For troubleshooting purposes DDC is well suited for the following scenarios.

### 4.1 Non Linear Convertor Card Issues

Ratio of scans. Works for air or bow-tie filters, but not a mix and match. See [Using DDC on page 33](#).

### 4.2 Beam Obstruction Issues

- Polaroid film on collimator works well.
  - Tube @ 6 o'clock.
  - Three Polaroid films directly on the Collimator face.
  - One stationary exposure, air, 80 kv, 10 mA 16 x 1.25mm, 0.1 seconds (make the technique as low and short as possible)
  - Develop films and look for white particles on the negative. The negative will show particles better than the film.
- Ratio of scans. Works for air or bow-tie filters, but not a mix and match.

### 4.3 Detector Profile Scanning

Good for open and shorts.

Single stationary air filter exposure. Use Scan Analysis MSD plot.

### 4.4 Manual Interconnect Test

Similar to DASTOOLS Interconnect test only much faster if you know what you are looking for. See [Figure 1-2 on page 9](#), and [Figure 1-3 on page 10](#) for FET modes and Protocol names.

### 4.5 Isolating Mechanical Noise

- Use progress noise scanning technique. See [2.4.2 - Using DDC](#).
- Use Scan Analysis MSD plot for evaluation.

## 4.6 Z FET Control Issues

- Use FET override modes to independently select Data and Z-Axis FET modes.
- Note: The first release M4 software contains Low Dose FET mode selections. These modes are not used in application scanning. Additionally the Z-Channels are from 763 through 768. The GUI is wrong. This should be corrected to reflect applications modes in the next software release.
  - Set-up a low technique Stationary Xray On scan for 30 seconds.
  - Select desired FET Override modes.
  - Accept protocol BUT DO NOT START SCAN
  - Look at backplane LEDS for FET mode Pattern.
  - DAS will default back to FET mode 54, idle state.

| DATA CHANNEL   | FET MODE | Z CHANNEL            | FET MODE |
|----------------|----------|----------------------|----------|
| 4 x 0.625_LD   | 62       | 4 x 0.625_LD_Z_MODE  | 0        |
| 4 X 1.25_LD    | 10       | 4 X 1.25_LD_Z_MODE   | 0        |
| 4 X 3.75       | 7        | 4 X 3.75_Z_MODE      | 48       |
| 8 X 1.25_LD    | 2        | 8 X 1.25_LD_Z_MODE   | 39       |
| 8 X 2.5        | 1        | 8 X 2.5_Z_MODE       | 49       |
| 16 X 0.625_LD  | 6        | 16 X 0.625_LD_Z_MODE | 39       |
| 16 X 1.25      | 54       | 16 X 1.25_Z_MODE     | 49       |
| Z_SLOPE_CAL_1  | 4        | Z_SLOPE_CAL_1        | 4        |
| Z_SLOPE_CAL_2  | 40       | Z_SLOPE_CAL_2        | 40       |
| Z_SLOPE_CAL_3  | 41       | Z_SLOPE_CAL_3        | 41       |
| Z_SLOPE_CAL_4  | 42       | Z_SLOPE_CAL_4        | 42       |
| Z_SLOPE_CAL_5  | 43       | Z_SLOPE_CAL_5        | 43       |
| Z_SLOPE_CAL_6  | 44       | Z_SLOPE_CAL_6        | 44       |
| Z_SLOPE_CAL_7  | 45       | Z_SLOPE_CAL_7        | 45       |
| Z_SLOPE_CAL_8  | 46       | Z_SLOPE_CAL_8        | 46       |
| Z_SLOPE_CAL_9  | 47       | Z_SLOPE_CAL_9        | 47       |
| Z_SLOPE_CAL_10 | 48       | Z_SLOPE_CAL_10       | 48       |
| Z_SLOPE_CAL_11 | 49       | Z_SLOPE_CAL_11       | 49       |
| Z_SLOPE_CAL_12 | 50       | Z_SLOPE_CAL_12       | 50       |
| TRK3           | 39       | TRK3                 | 39       |
| ALL ZEROS      | 0        | ALL ZEROS            | 0        |
| ALL ONES       | 63       | ALL ONES             | 63       |

**Table 2-2 DDC FET Mode Overrides ONLY**

## Section 5.0

### QIF Visualization

See appropriate Advanced System Service manual for detailed information.

Words of wisdom.

- QIF Visualization Tool is not an artifact troubleshooting tool.
- QIF Visualization Tool is a Manufacturing Specification Visualization tool.
- QIF Visualization Tool and Scan Analysis used together make a powerful tool set.

## Section 6.0

### Power Supplies

Power supplies are responsible for a large portion of Image Quality and system performance issues. LightSpeed 16 introduces a failure mode that is difficult to isolate, EMI induced noise.

EMI induced noise has many root causes which will be covered later. Now it is important to understand that EMI noise can be seen only with a spectrum analyzer. The common oscilloscope and DVM are inadequate to identify or confirm this failure mode. Since the spectrum analyzer test equipment is not standard issue, we need to use and understand the software tools available to us.

#### 6.1 EMI Noise Defined

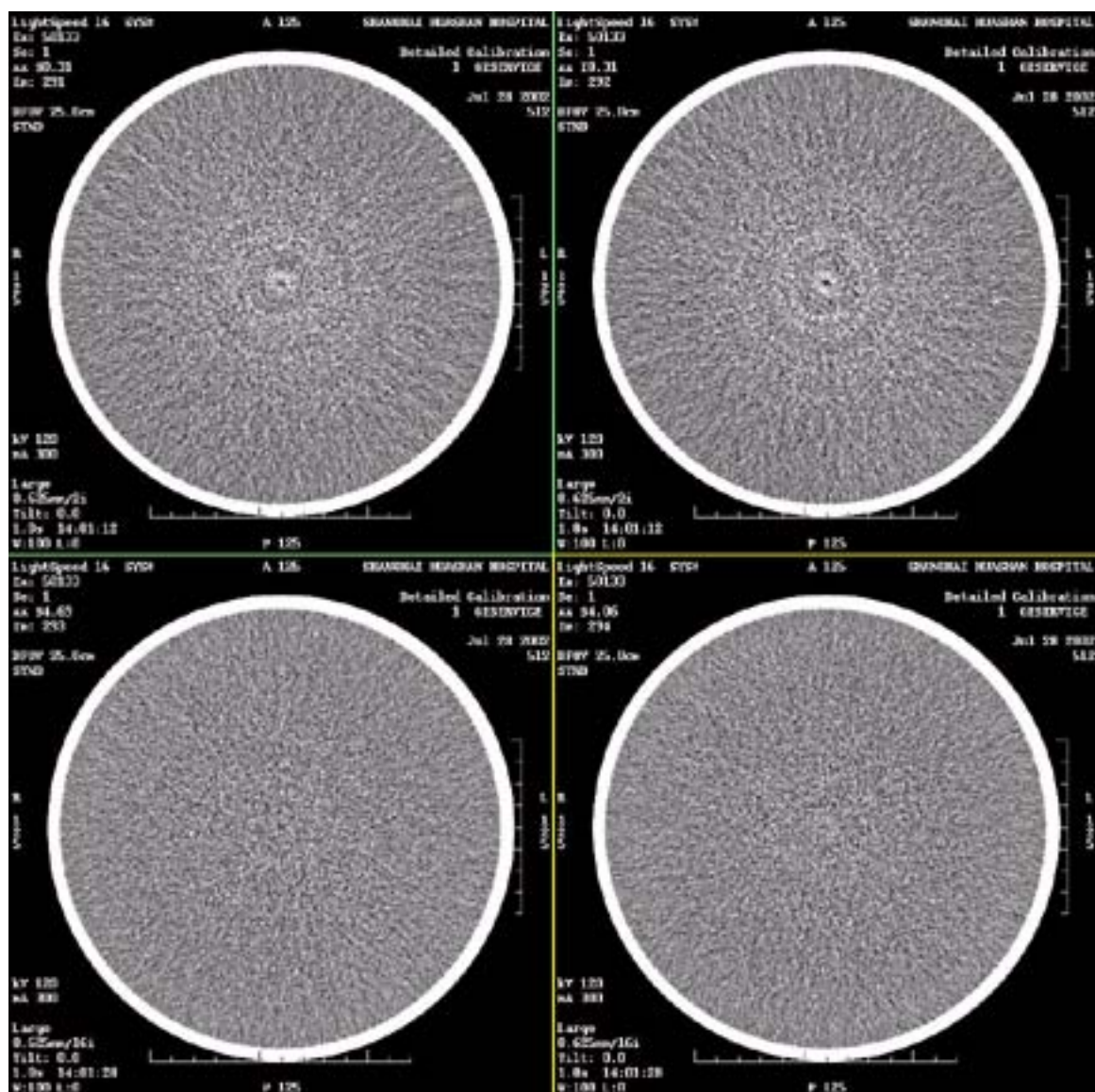
The DAS power supply package used is the source of the EMI noise. Specifically the line transformer used to step down the 120 vac. This is true primarily of the 5 volt analog supplies and to a lesser extent the 12 volt analog supplies. The 5 volt logic and 24 volt switching supplies do not exhibit this failure mode.

The core of the transformer saturates when the current load of the secondary becomes unbalanced. The power supply circuitry used to convert AC into DC is very efficient and successfully compensates for the unbalanced current load. The resultant power supply output DC waveforms generally appear clean with no significant AC components. This is why the oscilloscope and DVM are mostly ineffective in isolating the faulty power supply.

In general there are 3 basic reasons this can occur.

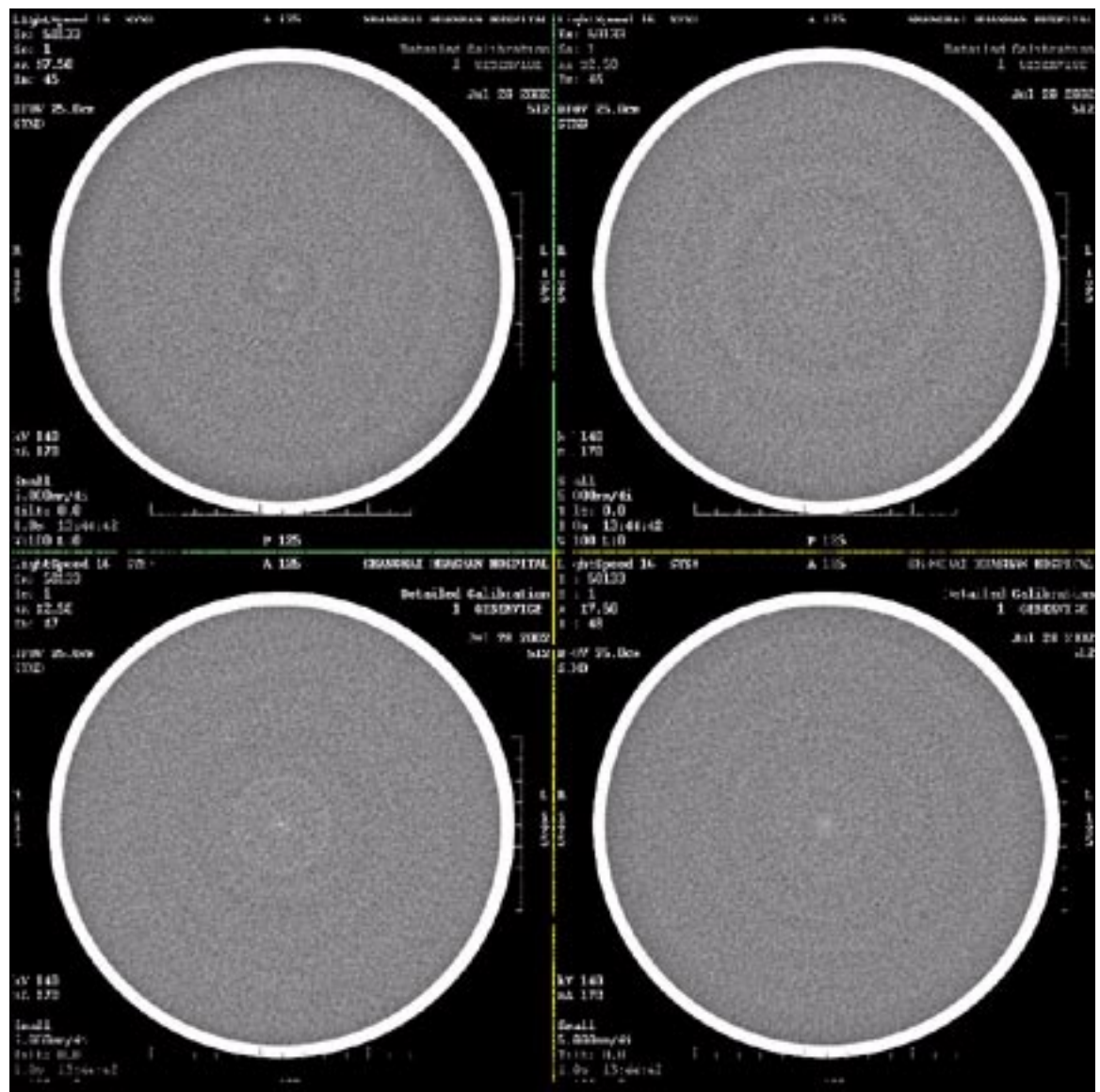
- 1.) Power supply quality.
  - Component failure.
  - Poor solder joints.
  - Workmanship assembly quality.
- 2.) Power distribution harness faults.
  - Poor crimps.
  - Pushed pins.
  - Damaged wires resulting in shorts or opens.
- 3.) Load faults
  - DAS chassis and associated component failures.

## 6.2 Identifying EMI Noise Issues



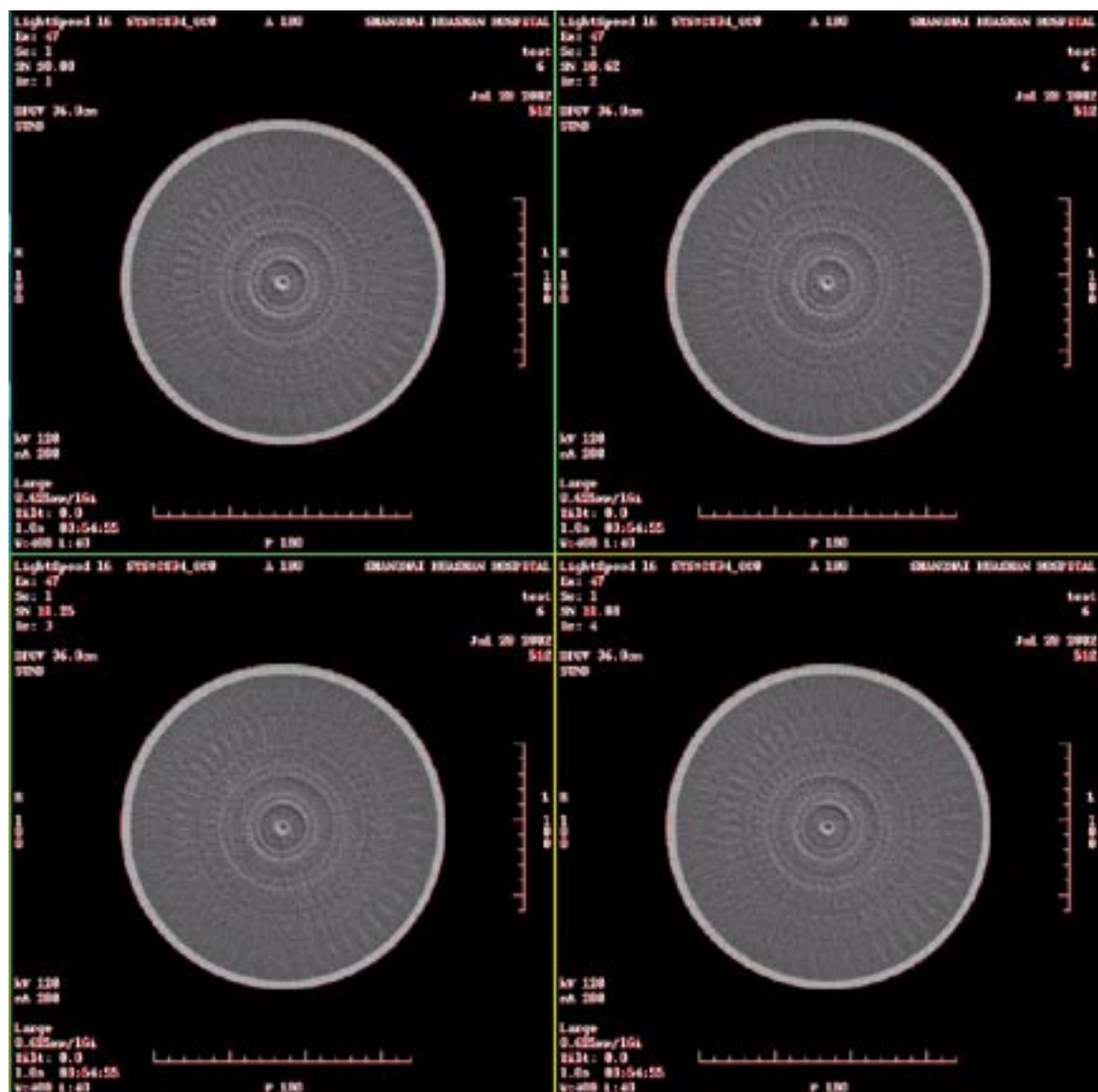
**Figure 2-3 Center Spot Artifact**

- Center Artifact after full calibrations.
- Elevated noise.
- CT Number shift from image to image.
- Root cause, +5 vdc Analog power supply. [See Figure 2-11 on page 50.](#)



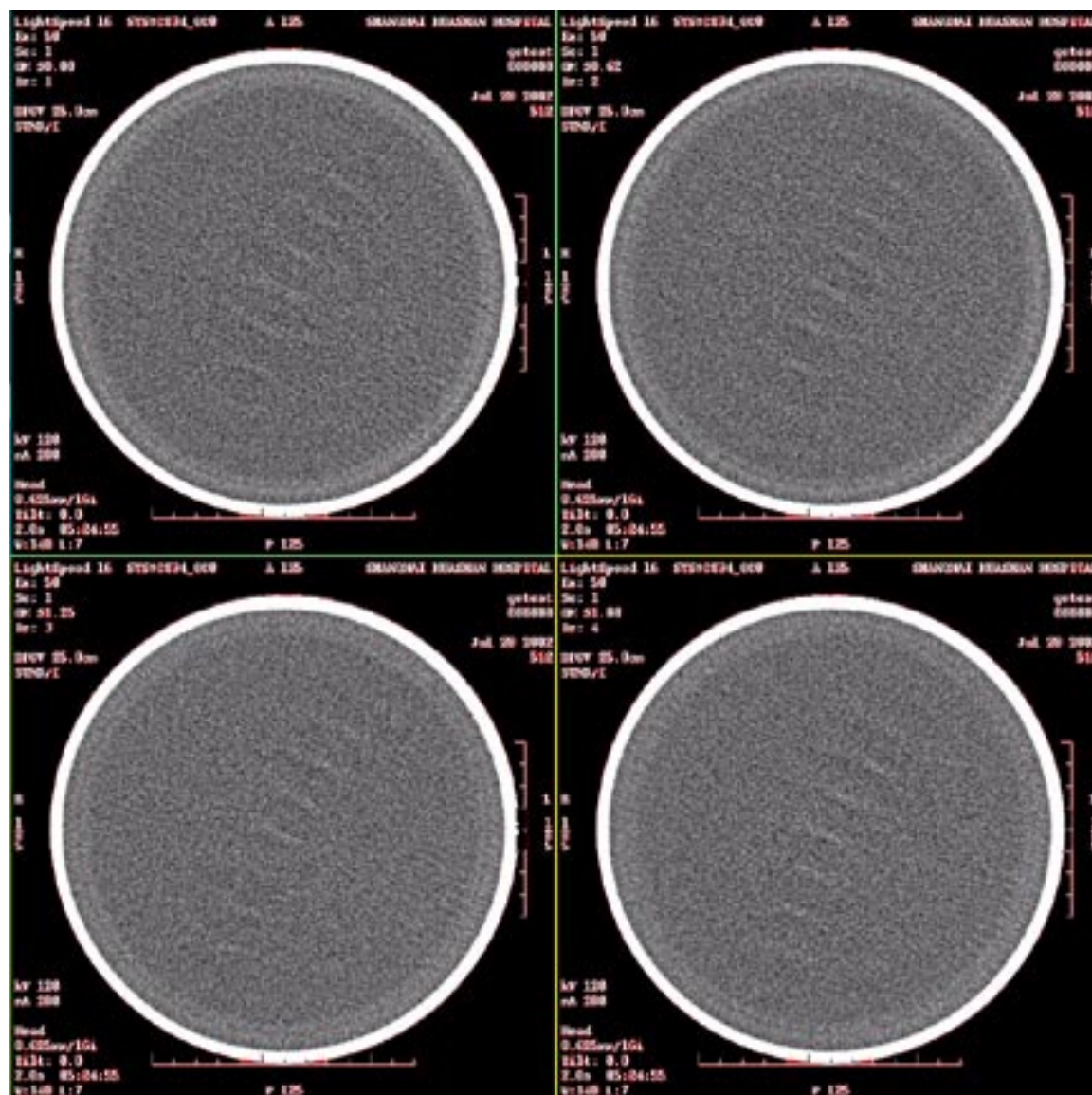
**Figure 2-4 Subtle Band Artifact**

- Same system as [Figure 2-3 on page 42](#), different protocol.
- Center artifact with subtle banding.
- Banding does not translate to converter card or detector module patterns.
- Root cause, +5 vdc Analog power supply.



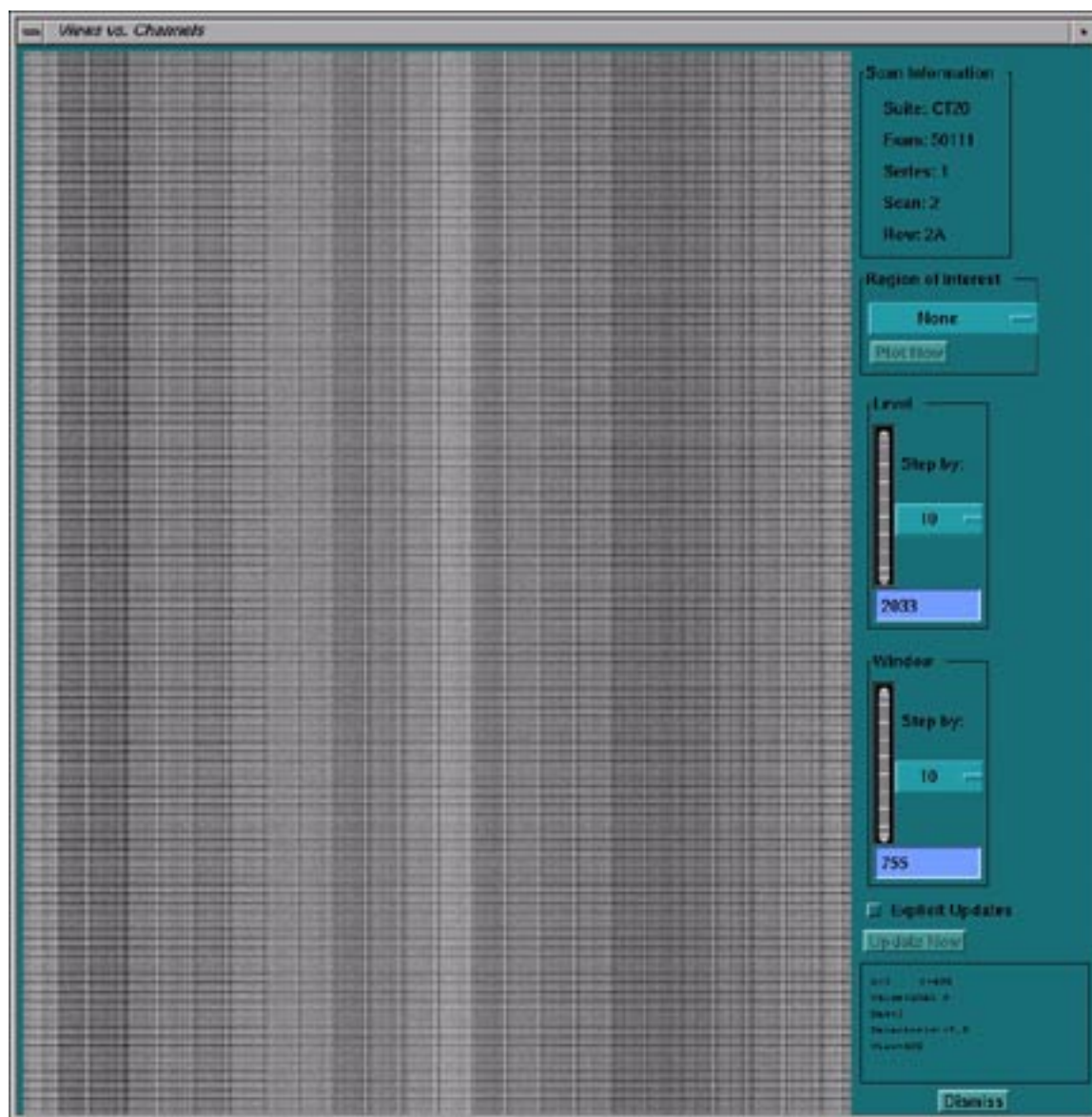
**Figure 2-5 Cornea (Eyeball) Artifact**

- Same system as previous 2 pages, different protocol.
- Center artifact with semi-hard rings.
- Radial pattern is across many channels.
- Root cause, +5 vdc Analog power supply. [See Figure 2-7 on page 46.](#)



**Figure 2-6 Eyebrow Artifact**

- Same system as previous 3 pages, different protocol.
- Soft partial banding with repeating pattern.
- CT number shift from image to image. [See Figure 2-11 on page 50.](#)
- Root cause, +5 vdc Analog power supply. [See Figure 2-8 on page 47.](#)



**Figure 2-7 VVC Analysis Detailed Cal**

- Artifact visible, repeating pattern in both views and channels.
- Pattern repeats approximately every 8 views.
- Channel pattern over several hundred channels can be seen.

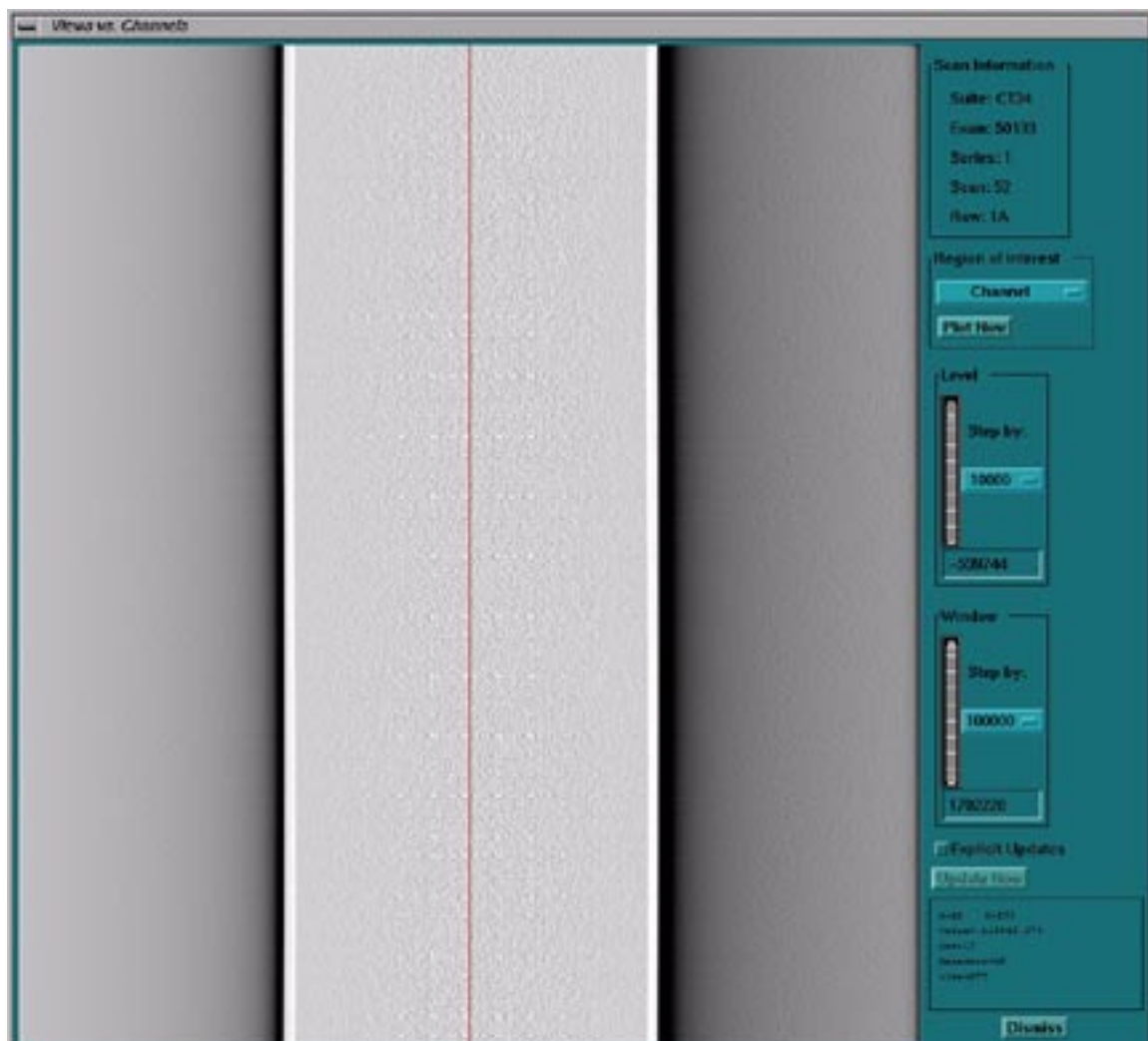
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### ***Conclusions***

Problem is most likely not internal to the DAS.

Problem is not Detector related.

Root cause effects entire DAS across entire exposure.



**Figure 2-8 VVC Analysis QA Phantom**

- Artifact visible, repeating pattern in both views and channels.
- Pattern repeats approximately every 8 views.
- Channel pattern repeats over several hundred channels, inside and outside of the phantom.

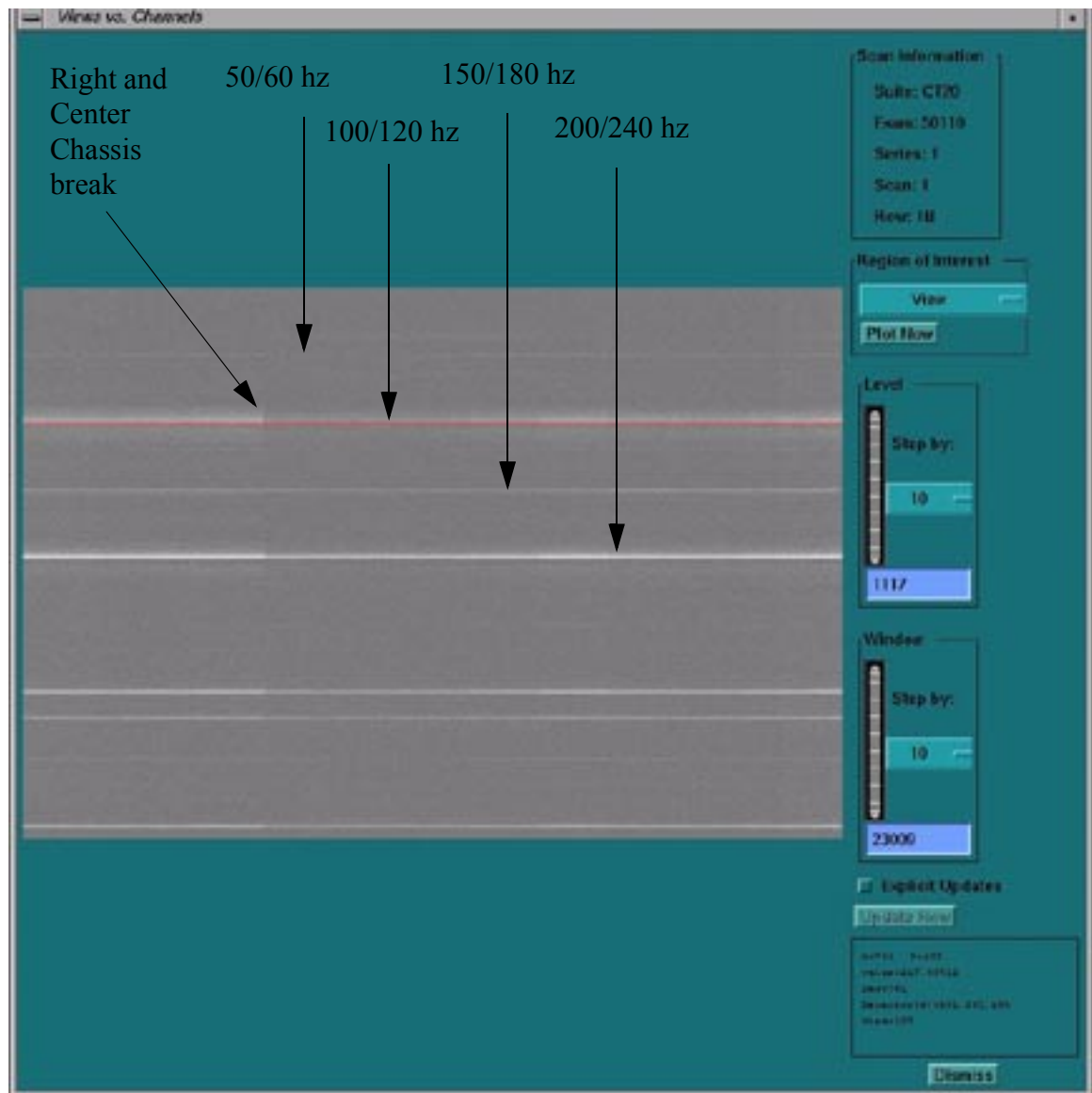
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### ***Conclusions***

Problem is most likely not internal to the DAS.

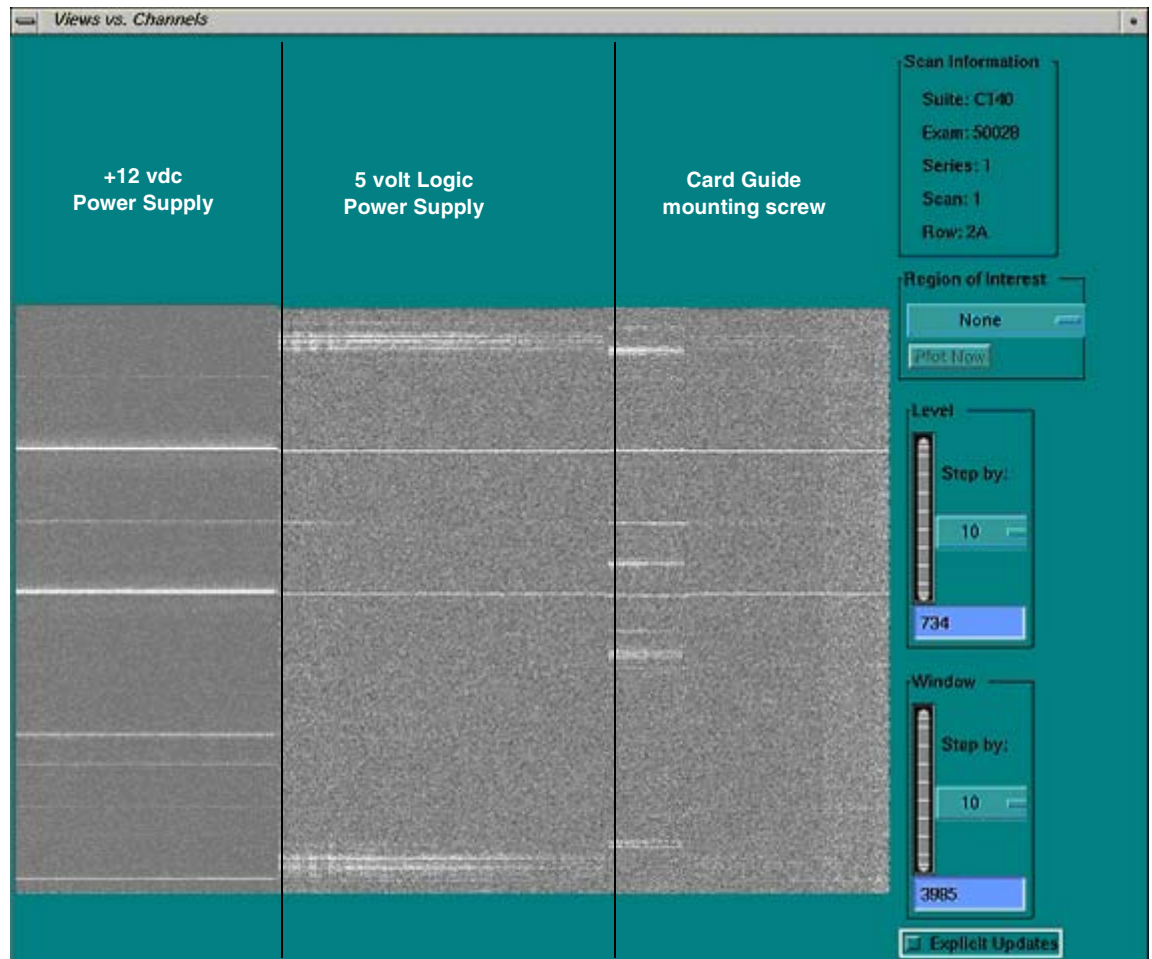
Problem is not Detector related.

Root cause effects entire DAS across entire exposure.



**Figure 2-9 FFT Analysis**

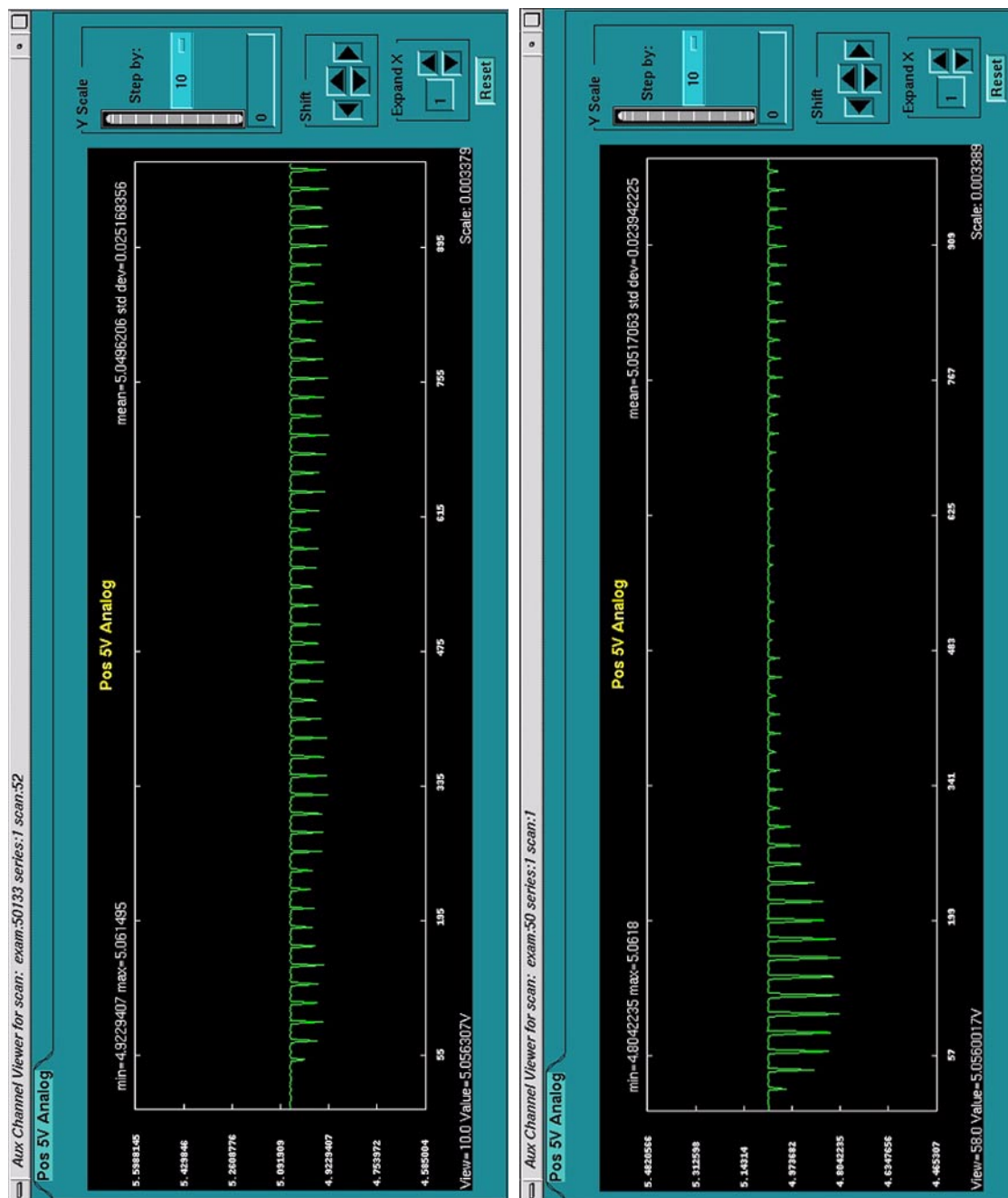
- Frequency versus Channel Plot
- Interpretation is subjective.
- Three white horizontal lines are normal.
  - 50/60 hz, 100/120 hz, and 200/240 hz are generally normal.  
Additional lines indicate a noise source present at that frequency or frequency band. These can be harmonics of the original noise source.
  - Intensity determines if issue is present.
- User can plot using "Plot Now" and "Region of Interest" functions for detailed analysis.



**Figure 2-10 FFT Composite Example**

Figure 2-10 is not a real screen capture. It has been digitally manipulated to show 3 distinct failures observed in manufacturing.

- +12 vdc power supply
  - Using the cursor you would see a horizontal white line at approximately view 60, 120, 180 and 240. These “views” are the frequencies of the noise. The top 4 lines have a large intensity.
  - Use Aux channels and Z-Axis Channels to look for additional less subjective evidence.
  - Using an oscilloscope may not show any issue if it is an EMI failure. A spectrum analyzer will show these frequencies, but an amplitude for go no-no threshold has not been established.
- 5 volt logic power supply
  - A noise burst is seen across a band of frequencies for about 1/3 of the data collection.
  - A mirror image also appears at the bottom of the plot.
  - Use Aux Channels to view the plot to help verify fault.
- Card guide mounting screw
  - The DAS chassis card guides are secured with two button head cap screws. Each screw also uses a flat washer. The flat washer can extend into the channel that the convertor card slides in. This generates microphonics noise.
  - Screws are located about 2 cards from each chassis end on both top and bottom rails.
  - Remove or reposition washer and retest.



**Figure 2-11 Auxiliary Channels +5vdc Analog 50 hz**

- Both plots are from the same power supply.
- Negative spikes are every 20 views.
  - 984 views, 1 second scan > each view  $\approx$  1 msec
  - 20 views = 20 msec
  - $1 / 0.020 = 50$  hz
- Notice sinusoidal pattern of EMI noise spikes. This is the CT number shift.

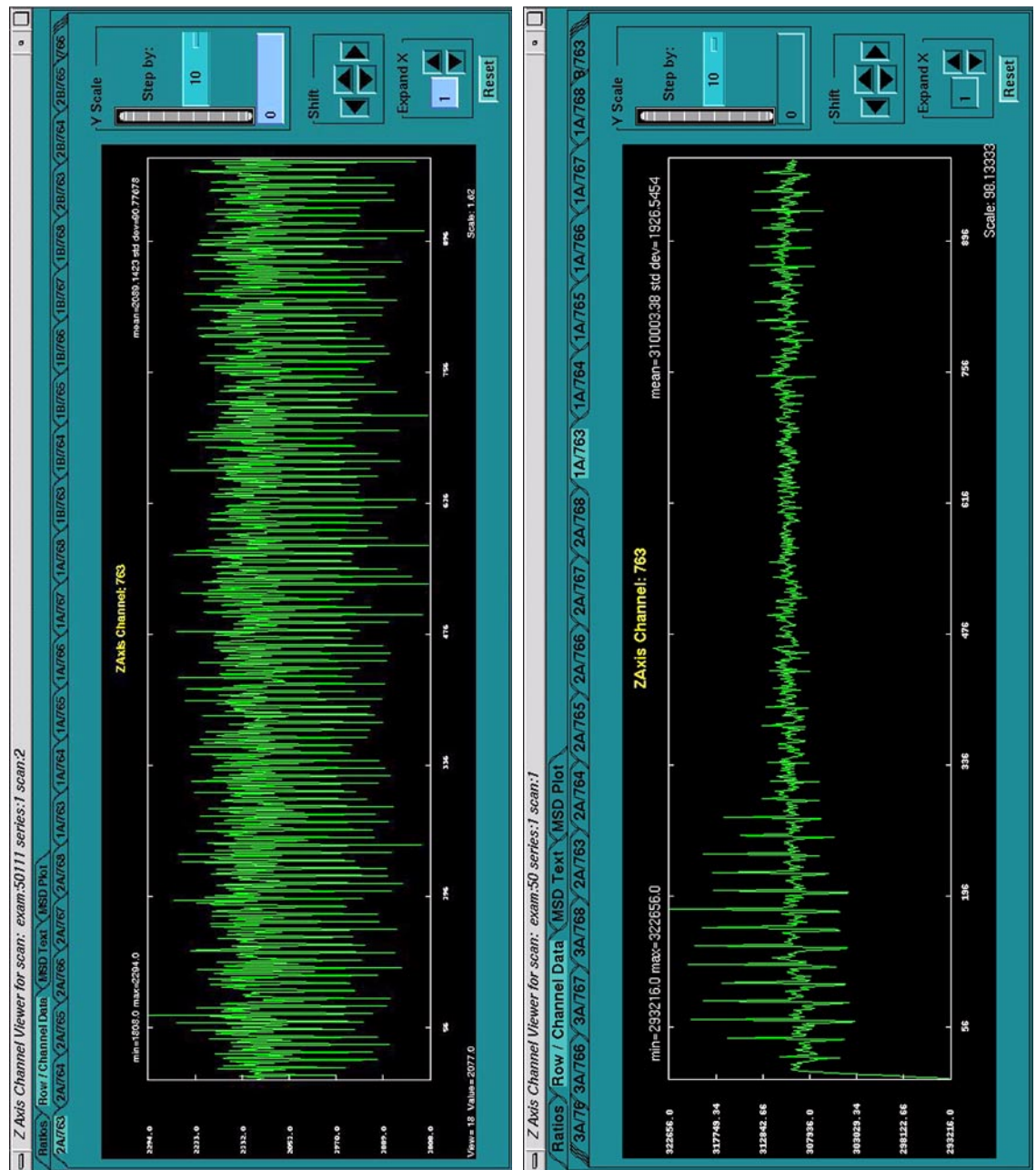
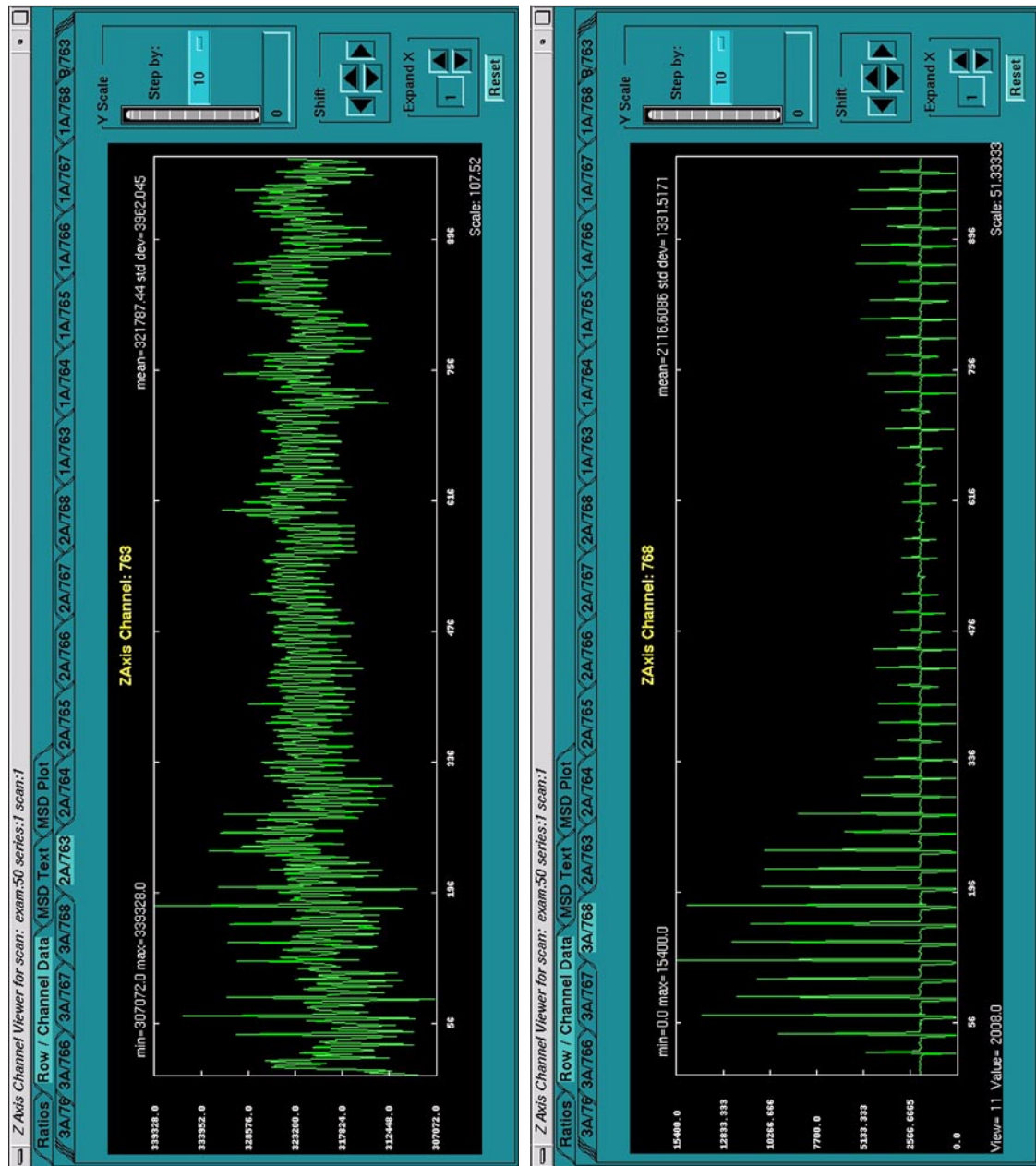


Figure 2-12 Z-Axis Channels +5vdc 120 hz (Left) 50 Hz (Right)

- Z-Channel noise, left plot has negative spikes every 8 views.  
 $1 / 0.008 = 125 \text{ hz}$
- Z-Channel noise, right plot has positive spikes every 20 views for approx 300 views.  
 $1 / 0.020 = 50 \text{ hz}$
- Compare 50 hz noise plot with [Figure 2-11 on page 50](#) right plot, and both plots in [Figure 2-13 on page 52](#).



**Figure 2-13 Z-Axis Channels +5vdc 50 Hz again**

Compare right plot of [Figure 2-11 on page 50](#), right plot of [Figure 2-12 on page 51](#) and both plots of [Figure 2-13](#).

- This is the same Exam, Series, and Image. EMI noise source is +5 vdc analog power supply.
- EMI field intensity effects LEFT DAS only. Flux lines intersect specific sections of converter cards.
  - Net result is Z-Axis channel distortion differs across each macro row.
  - This is a POSITIVE DAS Power supply EMI failure signature.

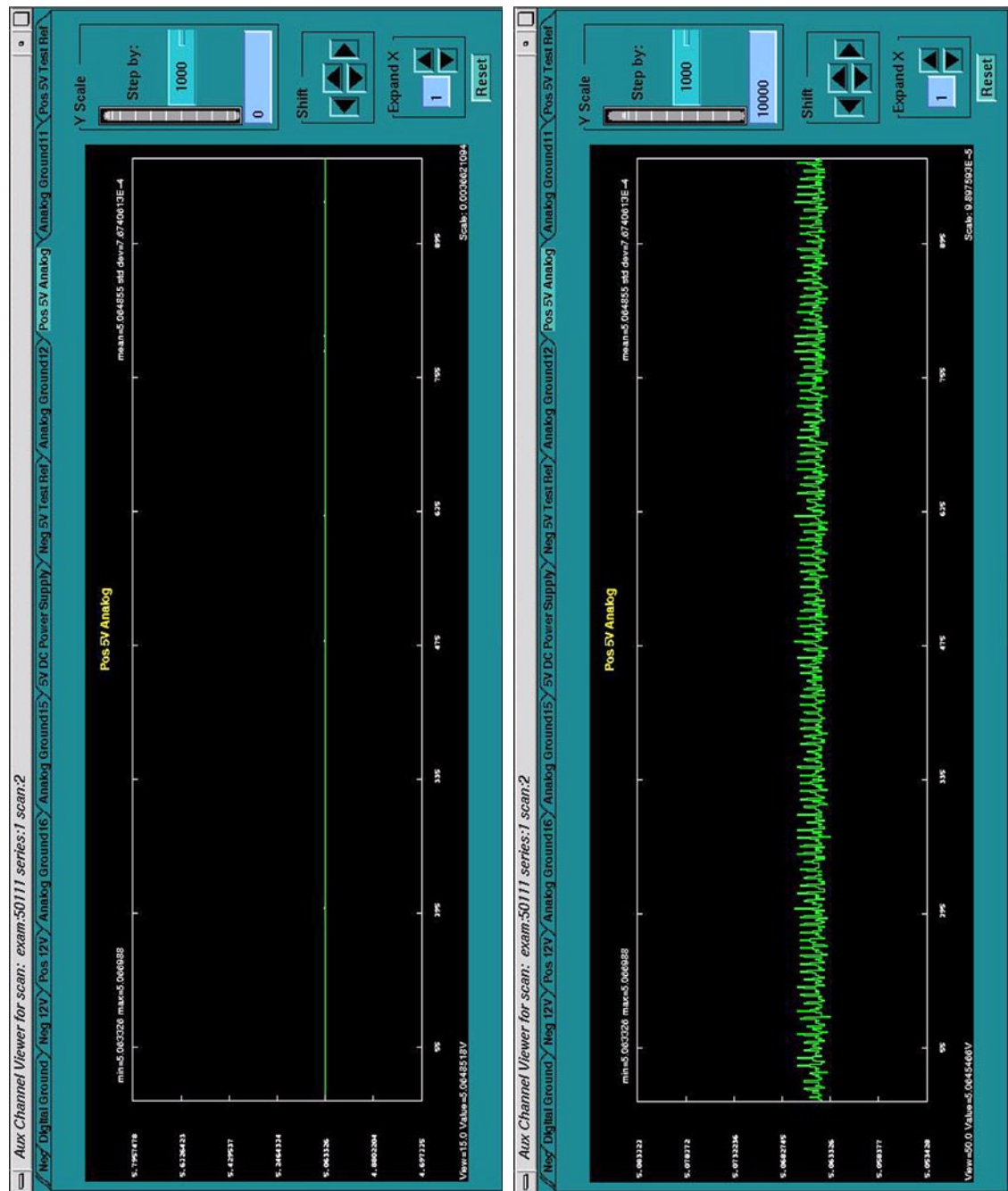
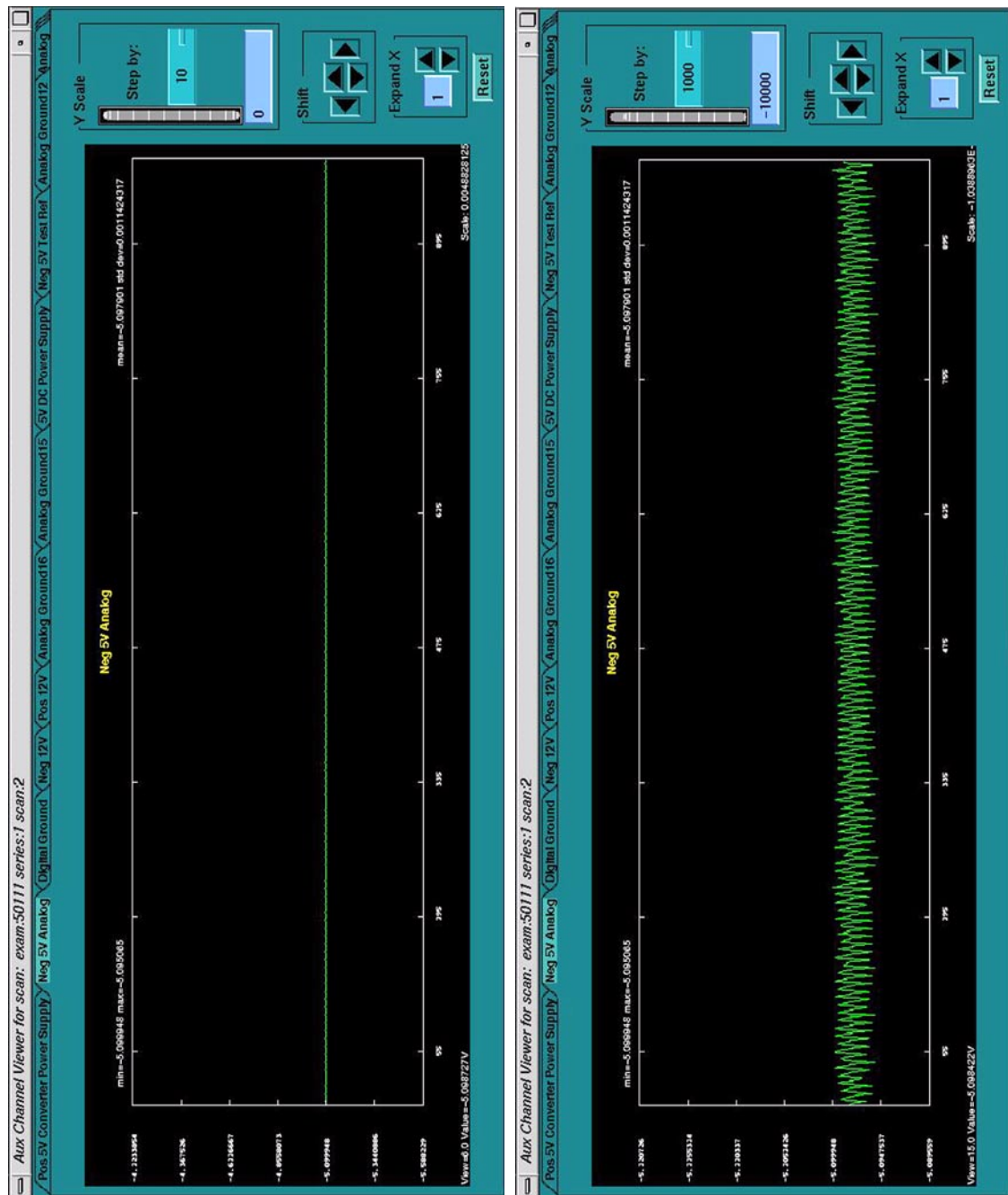


Figure 2-14 Auxiliary Channels +5vdc Analog

- Notice small bumps on left plot.
  - Normal plot is a flat line.
  - Small bumps indicate potential EMI issue.
- Right plot shows positive spikes every 8 channels.  
 $1 / 0.008 = 125 \text{ hz}$

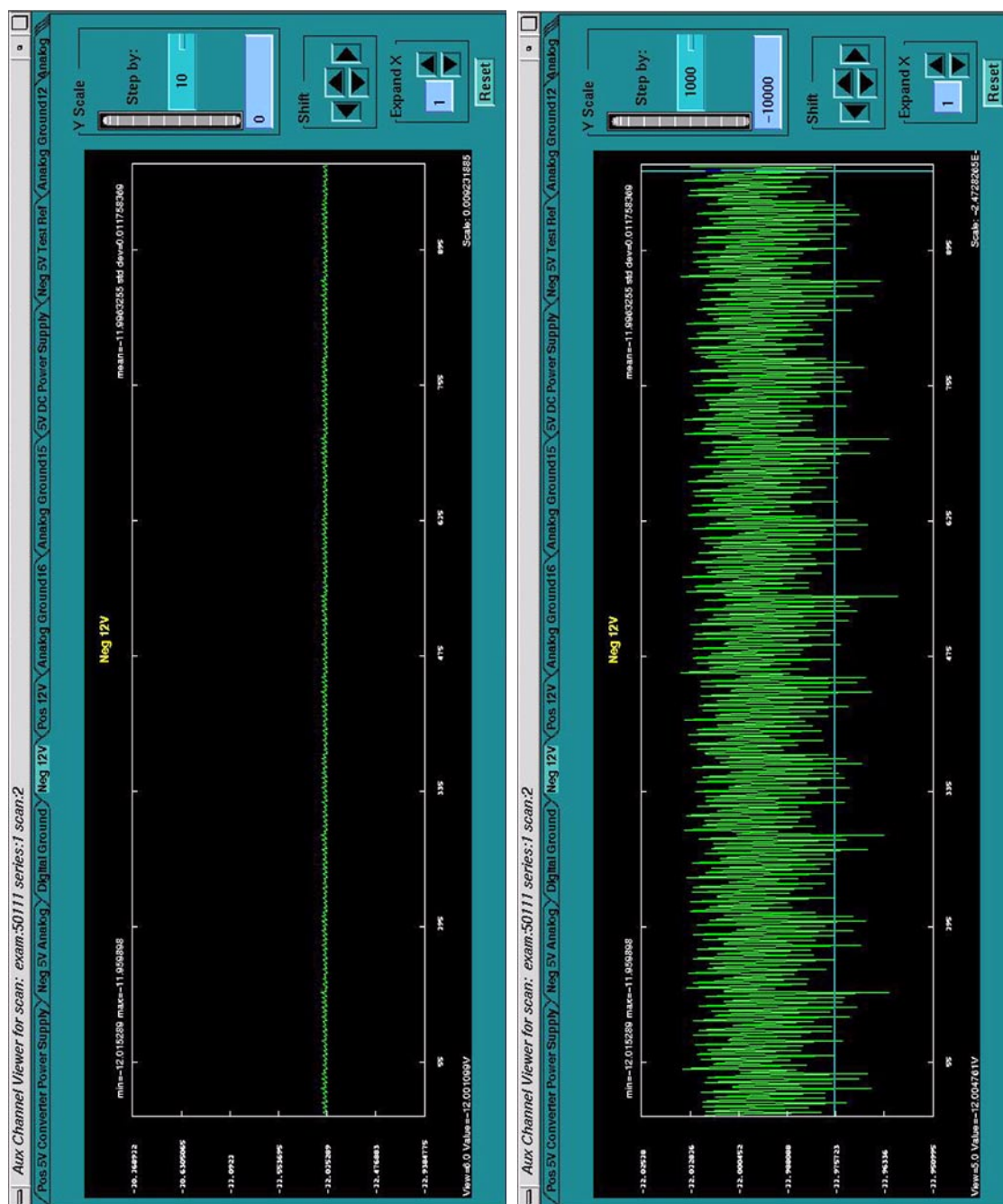
This caused DASTOOLS Noise and Offsets failures and Image Quality issues.



**Figure 2-15 Auxiliary Channels -5vdc Analog**

- Notice small bumps on left plot.
  - Normal plot is a flat line.
  - Small bumps indicate potential EMI issue.
- Right plot shows negative spikes every 8 channels.
  - $1 / 0.008 = 125 \text{ hz}$

This caused DASTOOLS Noise and Offsets failures and Image Quality issues.



**Figure 2-16 Auxiliary Channels -12vdc Analog**

- Notice small bumps on left plot.
  - THIS IS NORMAL FOR THE 12 volt analog power supplies
  - Magnify Y Scale to 10000
- Right plot shows
  - Random negative spiking of increasing amplitude.
  - Largest negative spikes are approx 85 views apart. (About 12 hz)

No Image Quality or DASTOOLS issues from this situation.

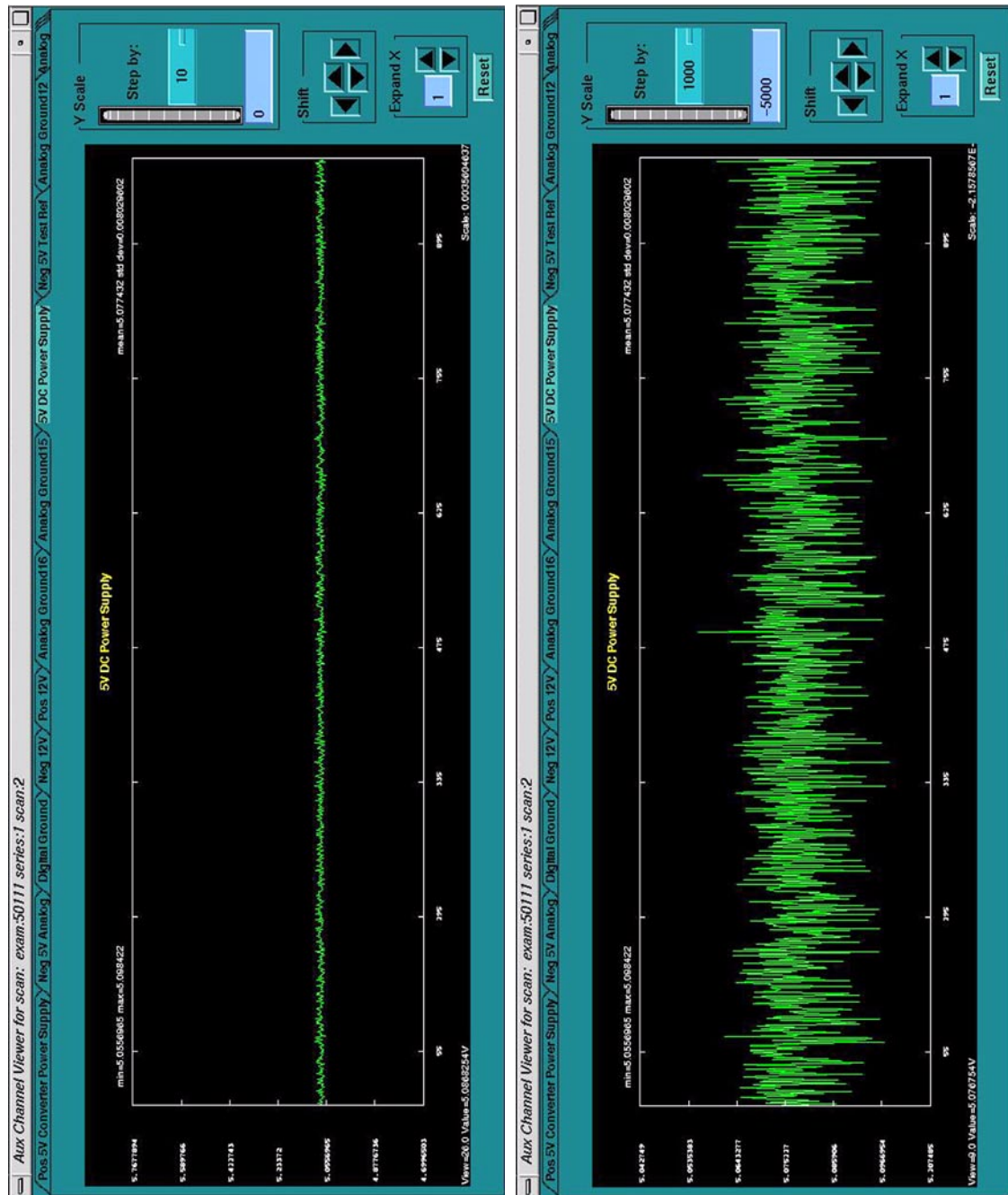


Figure 2-17 Auxiliary Channels 5vdc Logic

- Notice small bumps on left plot.
  - THIS IS NORMAL FOR THE 5 volt logic power supply.
- Right plot shows
  - Random spiking.

No Image Quality or DASTOOLS issues from this situation.

### 6.2.1 EMI Noise Summary

Any EMI issue can cause the following in any combination.

- Image Artifacts that can change from one exam to another.
- Image Artifacts that can change from series to series.
- CT number shift from Image to Image without artifacts.
- DASTOOLS Noise and Offsets failures.
- DAS initialization failures

Positive 5 and 12 volt power supplies induce noise in the left DAS chassis.

Z-Axis channels can be severely distorted.

Negative 5 and 12 volt power supplies induce noise in the right and center DAS chassis.

Power supply transformer location determines induced noise within the DAS.

## 6.3 Power Supply Troubleshooting

### 6.3.1 5 Volt Analog

| SYMPTOM                                                                                               | POSSIBLE CAUSE                                                | CORRECTIVE ACTION                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Voltage at TP5 OR TP8, (+/- 5V_A) to TP4 (AGND) is incorrect.                                         | Maladjusted voltage pot                                       | Adjust "Voltage Adjust" pot on the supply.                                                                                                                                    |
| Voltage at TP5 OR TP8, (+/- 5V_A) to TP4 (AGND) is low and cannot be adjusted to the correct voltage. | High resistance between power supplies and center backplane   | Inspect solder connections to power supply output terminals; re-solder as required.                                                                                           |
|                                                                                                       |                                                               | Verify J96 connection is tight.<br>Inspect DC Mate-N-Lok connectors, look for pushed pins, bad crimps, bent or broken pins; repair or replace as required.                    |
|                                                                                                       | Defective power supply                                        | Replace power supply if all other troubleshooting steps fail.                                                                                                                 |
| Voltage at TP5 OR TP8, (+/- 5V_A) to TP4 (AGND) is too high, and cannot be adjusted correctly.        | Floating DC return.                                           | Inspect DC Mate-N-Lok connectors, look for pushed pins, bad crimps, bent or broken pins; repair or replace as required.                                                       |
|                                                                                                       | Defective power supply                                        | Replace power supply.                                                                                                                                                         |
| Voltage at TP5 OR TP8, (+/- 5V_A) to TP4 (AGND) is 0V                                                 | No 120VAC power to the power supply                           | Verify fuse box toggle switch is turned on.<br>Check +/-5V_A fuses in fuse box, replace fuse if necessary.                                                                    |
|                                                                                                       | Wiring defect                                                 | J96 connector/harness continuity using DVM.<br>Inspect AC and DC Mate-N-Lok connectors, look for pushed pins, bad crimps, bent or broken pins; repair or replace as required. |
|                                                                                                       |                                                               | Inspect solder connections at transformer input lugs; re-solder if necessary.                                                                                                 |
|                                                                                                       | Defective power supply                                        | Replace power supply if all other troubleshooting steps fail.                                                                                                                 |
| Excess DAS noise without xray, especially in higher channels at Gain 31                               | Defective power supply                                        | First rule out converter cards or detector, then replace power supply.                                                                                                        |
| Output voltage is zero or near zero                                                                   | Current limit pot was adjusted instead of voltage adjust pot. | This should not occur. The current limit pot has glyptol on it to prevent field adjustment. This pot is OEM set. If disturbed the power supply should be replaced.            |

**Table 2-3 DAS 5 Volt Analog Power Supply Troubleshooting**

## 6.3.2 12 Volt Analog

| SYMPTOM                                                                                              | POSSIBLE CAUSE                                                                          | CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Voltage at TP3 or TP6 (+/- 12V_A) to TP4 (AGND) is incorrect                                         | Maladjusted voltage pot                                                                 | Adjust "Voltage Adjust" pot on the Master supply.                                                                                                                                                                                                                                                                                                           |
| Voltage at TP3 or TP6 (+/- 12V_A) to TP4 (AGND) is low and cannot be adjusted to the correct voltage | Master/Slave current share jumper is not connected or open. See Figure 2-18 on page 60. | Verify the jumper between the two supplies is connected, no pushed pins, bad crimps, bent or broken pins, and connectors are wired correctly; correct as necessary.                                                                                                                                                                                         |
|                                                                                                      |                                                                                         | Verify solder connection of jumper wire to EXT_DRV terminals on power supplies; re-solder as required.                                                                                                                                                                                                                                                      |
|                                                                                                      | High resistance between power supplies and center backplane                             | Inspect solder connections to power supply output terminals; re-solder as required                                                                                                                                                                                                                                                                          |
|                                                                                                      |                                                                                         | Verify J96 connection is tight.<br>Inspect DC Mate-N-Lok connectors, look for pushed pins, bad crimps, bent or broken pins; repair or replace as required.                                                                                                                                                                                                  |
|                                                                                                      | Defective power supply                                                                  | Disconnect both 12V DC Mate-N-Lok connectors. Disconnect the current share jumper. Apply power, measure the voltages at connectors. The "Master" output should be 12.0-12.5V. The "Slave" output should be 11.0 – 12.0V.<br>Replace defective supply as required.                                                                                           |
| Voltage at TP3 or TP6 (+/- 12V_A) to TP4 (AGND) is low and cannot be adjusted to the correct voltage | Defective power supply (current limit)                                                  | Disconnect right DAS chassis power cable, J99. (Both supplies should pass this partial load test.)<br>Disconnect the "Master" DC output Mate-N-Lok. Verify 11 volts at TP3 or TP6, (slave output).<br>Power down, connect "Master" DC output and disconnect "Slave" DC output. Verify 12 volts at TP3 or TP6, (master output).<br>Replace defective supply. |
| Output voltage is zero or near zero                                                                  | Current limit pot was adjusted instead of voltage adjust pot.                           | This should not occur. The current limit pot has glyptol on it to prevent field adjustment. This pot is OEM set. If disturbed the power supply should be replaced.                                                                                                                                                                                          |
| Voltage at TP3 or TP6 (+/- 12V_A) to TP4 (AGND) is 0V                                                | No 120VAC power to the power supply                                                     | Verify fuse box toggle switch is turned on.<br>Check +/-12V_A fuse in fuse box, replace fuse if necessary                                                                                                                                                                                                                                                   |
|                                                                                                      | Wiring defect                                                                           | J96 connector/harness continuity using DVM.<br>Inspect AC and DC Mate-N-Lok connectors, look for pushed pins, bad crimps, bent or broken pins; repair or replace as required.                                                                                                                                                                               |

Table 2-4 DAS 12 Volt Analog Power Supply Troubleshooting

| SYMPTOM | POSSIBLE CAUSE         | CORRECTIVE ACTION                                                                                   |
|---------|------------------------|-----------------------------------------------------------------------------------------------------|
|         |                        | Inspect solder connections of wires soldered to the transformer input lugs; re-solder if necessary. |
|         | Defective power supply | Replace power supply if all other troubleshooting steps fail.                                       |

Table 2-4 DAS 12 Volt Analog Power Supply Troubleshooting (Continued)

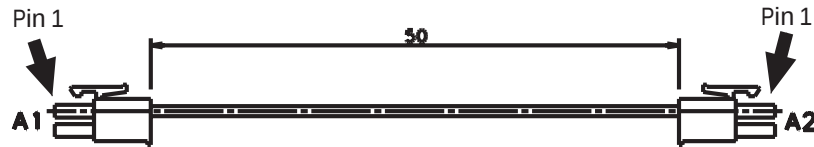


Figure 2-18 Current Share Jumper for +/- 12 vdc Power Supplies

### 6.3.3 5 Volt Logic

| SYMPTOM                                                                               | POSSIBLE CAUSE                                                | CORRECTIVE ACTION                                                                                                                                                  |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Voltage at TP1 (+5V_L) to TP2 (LGND) is incorrect                                     | Maladjusted voltage pot                                       | Adjust "Voltage Adjust" pot on the supply.                                                                                                                         |
|                                                                                       | Remote sense is jumpered out                                  | Verify power supply jumpers to S+ and S- are both removed                                                                                                          |
| Voltage at TP1 (+5V_L) to TP2 (LGND) is low and cannot be adjusted to correct voltage | High resistance between power supply and center backplane     | Verify screw terminal connections to the power supply are tight. Inspect for bad crimps.                                                                           |
|                                                                                       |                                                               | Verify J104 connection is tight.                                                                                                                                   |
|                                                                                       | +12V_A and/or -12V_A supplies are shut down                   | Measure the voltages at Center Backplane test points +12V_A to AGND, -12V_A to AGND. If these are <2V, investigate and correct.                                    |
|                                                                                       | Short circuit or overload in the +5V_L circuit                | Look for a short circuit in the +5V_L circuit. Find and correct short circuit as required.                                                                         |
| Output voltage is zero or near zero                                                   | Current limit pot was adjusted instead of voltage adjust pot. | This should not occur. The current limit pot has glyptol on it to prevent field adjustment. This pot is OEM set. If disturbed the power supply should be replaced. |
| Voltage at TP1 (+5V_L) to TP2 (LGND) is 0V                                            | No 120VAC power to the power supply                           | Verify fuse box power toggle switch is turned on.<br>Check +5V_L fuse on switch box, replace fuse if necessary                                                     |
|                                                                                       |                                                               | Inspect lugs connecting to the power supply AC input terminals for bad crimps or broken lugs; repair or replace as require.                                        |
|                                                                                       | Defective power supply                                        | Replace power supply if all other troubleshooting steps fail.                                                                                                      |

Table 2-5 DAS 5 Volt logic Power Supply Troubleshooting

## 6.4 Manufacturing Issues Identified

| DAS INTEGRATION ISSUE                                                     | SYMPTOMS                                                                                                                                                           | POSSIBLE ROOT CAUSE                                                                                                                                                         | SOLUTION                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zero ohm resistors on Center backplane blown. (R2,R3,R4).                 | DCB aux channel reads 7V for the 5V supply                                                                                                                         | Power supply shorted to chassis or a power cable short. Power transistors or rectifiers are missing an insulating pad.                                                      | Ohm out the transistor case to ground. It should not connect. Ohm out power cables for shorts to grounds or chassis. Replace chassis, cables or power supply as required.                                                         |
| Left Zone Detector heater shorting to plenum                              | Left Zone heater gets the Detector too hot                                                                                                                         | The DAS Plenum air dam is touching the Left Detector heater thermostat.                                                                                                     |                                                                                                                                                                                                                                   |
| DAS Power supply errors.                                                  | Noisy 12V_A supplies or the supplies crowbar                                                                                                                       | 12V_A supply current sharing leads not connected or incorrect pin location in connector housings. Possible bad supply or cable. <a href="#">See Figure 2-18 on page 60.</a> | Visually inspect sharing connection & correct. Check connectivity of cables for shorts to shield or ground. Disconnect supplies from each other and measure output voltage of each individual supply. Replace defective supplies. |
| Boards failing DAS Integration Noise tests                                | Noise on convertor cards in slots 62,63,90,91,22,23 due to screw and washer for card guides                                                                        | A metal washer is used to support the head of the head screw on the card guides. This moves sideways and contacts the Convertor board.                                      | Remove washer from the screw.                                                                                                                                                                                                     |
| Boards failing DAS Integration Noise tests at Gain 31.                    | Use the QIF tool to identify location of microphonics. If they are on the left edge of the left column of ASICs on the map, it may be the card guide charging up). | The card guide may be charged up.                                                                                                                                           | Remove the Convertor card and use a de-ionizing fan to blow on the card guide. Also blow on the Convertor card. Re-insert the card while blowing in the slot with the fan.                                                        |
| DAS configuration not recognized or DAS does not successfully initialize. | Parity Errors                                                                                                                                                      | Pins on Ribbon cable header not soldered or shorted, Ribbon cable not fully inserted, ground tabs bent & shorting ribbon cable header pins.                                 | Visually inspect cable and header for pin problems and solder connections. Try a new cable.                                                                                                                                       |
|                                                                           | Parity Errors                                                                                                                                                      | Convertor cards' Xilinx doesn't see shift clock.                                                                                                                            |                                                                                                                                                                                                                                   |
|                                                                           | Convertor card LEDs blink 8 times                                                                                                                                  | Shift clock lost                                                                                                                                                            |                                                                                                                                                                                                                                   |
| A Convertor board has a problem with serial stream                        | Parity errors in addition to an error stating the Convertor board does not initialize. (Move this board first in troubleshooting.)                                 | Bad pECL driver or receiver on the Convertor board.                                                                                                                         | Replace Convertor board                                                                                                                                                                                                           |
| DAS fails interconnect test                                               | Large contiguous sections of the backplanes have interconnect failures                                                                                             | Cable open or short or FET lines shorted together.                                                                                                                          | Ribbon cable has a problem-Replace cable. If not the cables, isolate the problem to a chassis by disconnecting the cables and then troubleshoot the DDIF.                                                                         |
| Detector temperature errors                                               | Bad Detector heating (low), faults on Convertor cards, DCB resets.                                                                                                 | Shrink tubing too close to the connector shell. This prevents the full insertion of the pin in the housing.                                                                 | Pins pushed out of Mate-N-Lok connectors. Inspect cables and re-insert pins.                                                                                                                                                      |

Table 2-6 Manufacturing DAS Integration Issues

| DAS INTEGRATION ISSUE                                  | SYMPTOMS                                                                                                                                                                                       | POSSIBLE ROOT CAUSE                                                                                                                                                                                    | SOLUTION                                                                                                                                             |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detector temperature low errors                        | Center Detector zone temperature is low                                                                                                                                                        | Center Detector zone needs 30 minutes minimum time to warm up.                                                                                                                                         |                                                                                                                                                      |
| Detector temperature errors                            | Detector temperature errors                                                                                                                                                                    | Cable mis-wired in the Gantry harness                                                                                                                                                                  | Correct cable mis-wire.                                                                                                                              |
| DAS does not initialize correctly on power up.         | System reports Converter CAN errors                                                                                                                                                            | Divider is too flimsy and it bends. Not always assembled to the chassis either. Divider (or the gasket attached to it) between the DCB and Converter card shorts out the CAN bus.                      | Inspect divider and correct problem by removing and replacing the divider or adjusting it.                                                           |
| DAS fails noise test                                   | Slot 29 in Detector Rows 2A and 1A have high noise.                                                                                                                                            | Slot 29 is more susceptible to power supply noise. Noisy Analog supplies (60 Hz, 120 Hz - use VVC, FFT to look at noise characteristics). 5V_L supply interferes with the slot also.                   | Replace Analog power supplies after measuring noise. If broadband noise, connect the cable shield at the 5V_L supply.                                |
| DAS fails offset and noise tests                       | High Means and Standard deviations across the whole DAS                                                                                                                                        | Power supplies may be noisy or DIP is bad.                                                                                                                                                             | Once power supplies are eliminated as a problem, check the DIP.                                                                                      |
| DAS does not initialize correctly on power up.         | Converter card LEDs stay on and don't blink. Full and partial chassis affected.                                                                                                                | Converter card reset lines shorted on backplane. Check for particles. Rotate DAS upside down and tap to see if something comes loose. Short between Converter Card Reset and D_IN_EVN+ (Pins B4 & C4). | Ohm Converter Reset to every pin on data connector with and without data cable connected. Try to dislodge any particles. Otherwise replace backplane |
| DAS fails interconnect test or has problems in IQ test | Makes it appear that a set of modules has its -5V bias line shorted.                                                                                                                           | Bad Converter card. Converter card front end goes high. About 2V                                                                                                                                       | Pull cards associated with affected modules until fault clears. Replace faulty board(s)                                                              |
| No +5V_L but +/-12V_A and +/-5V_A are OK               | Measured short between +5V_L and CGND with power cable connected but not with connector removed (then saw pin broken)                                                                          | Logic Connector had J104-1 (+5V_A) bent over shorting to chassis ground                                                                                                                                | Replace Center Chassis                                                                                                                               |
| DAS fails noise test                                   | Noise was high for all DAS channels on all Rows. Plot VVC/FFT plot for any row, more than two frequency bands on all channels is excessive. Plot aux channels (power only) and check for noise | Noisy -5V_A supply (would happen for +5V_A supply as well)                                                                                                                                             | Replace power supply                                                                                                                                 |
| DAS fails noise test                                   | Slot 1 very high. Stands out in QIF with the body of the box plot elevated.                                                                                                                    | Suspected something touching board or the shield is not correct. Turned out that slot 1 had contamination of some sort.                                                                                | Empty out nearby cards, guard with a de-ionized towel and use AMAX spray to clean around slot 1 connector.                                           |
| DAS fails data collection                              | Fails all data collection modes with zero words received.                                                                                                                                      | DAS Data Out fiber connected to DCB Receive port during staging or troubleshooting.                                                                                                                    | Connect DAS Data Out Fiber cable to DCB Transmit port.                                                                                               |

Table 2-6 Manufacturing DAS Integration Issues (Continued)

## 6.5 Troubleshooting Processes

### 6.5.1 General Suggestions

Look for patterns and physical locations information. Use the QIF tool or scan analysis and the “eye chart”.

If the QIF tool hangs, try selecting the QIF button again in DASTools. If that doesn't clear the QIF window, open a Unix shell and type “ps -ef” and look for a Netscape process that has the words “gui” or “interface” in it. Kill that process.

Shorts between two channels will usually have one channel's means go low and the other side of the short go high. Use the mapping tool or the QIF tool to identify if the shorts are adjacent pins on the DDIF or the Converter Card. A short to ground will have only one pin going to zero.

When you insert a Converter card, if you can't push it all the way in with your thumb, don't use the ejectors. If you do, you risk bending connector pins.

Opens will cause a drop in means in one channel and have the adjacent two channels peak up a little when an Xray scan is taken. This is because the blocked diode “bleeds” into the adjacent channels. You may not see anything in standard deviations.

Use Scan analysis views-vs-channels to see what the power supply noise looks like. Strong, multiple 60 hz and harmonics may indicate a problem. Aux channel noise is usually less than 4 mV. In some cases Aux Channels does not show power supply noise, the sampling rate is not high enough to achieve sufficient resolution. Use the View vs Channels, FFT plots with offset correction to see the frequencies of the supplies.

Board problems generally do not cross over more than two detector rows except for the ends. (See the “eye chart”). This is true for detector output / Asic input (DDIF) problems.

If multiple boards have failures, look to see if they all go to the same module. Then look at the module pinout to see if the pins are close.

If your noise is “moving”, look to see if it is going to adjacent physical locations on the Converter cards or the modules. This may indicate debris contamination.

If you have a noisy card in the center of a DAS and it is consistent, move it to the outside channels where the specs are relaxed. This can buy you time until the board is replaced.

If you use Amax, give it time to dry. If you spray into the connectors and cover them, they have stayed noisy in the past.

The center Detector heater takes about 30 minutes to come up to temperature (36°C) from a cold start. Keep this in mind if you get “detector temperature low” errors.

Magnifying glasses are useful. 99 cents will buy you a x2 pocket magnifier which works pretty good when looking at DDIF and connectors.

You can cross adjacent flex modules on the “A” side. This will narrow the search to something on the flex BGA or Detector.



#### NOTICE

**Do not connect detector flexes from the “A” side to the “B” side or the other way around. You will damage the detector FETs and/or backplane requiring replacement of both.**

An A/D failure on a converter card can cause very strange failures. The following has been observed:

- Fault 8 boards wide on every other board totaling 16 boards. (Can total 8 bds also).
- Fault seen on all macro rows across the detector, rows 8A through 8B.
- Detailed analysis shows bell curve distribution with failed board having largest peaks in means and zero data channels. Adjacent boards have large peaks with no zero data channels. Peak amplitude goes down the farther away you look from the failed board.

## 6.5.2 DDIF Specifics

Tools:

DAS Tools, QIF visualization tool up and running, Scan Analysis

Wrist strap, wrenches, specialized interposer and flex removal tools.

The QIF tool will help you identify board and module patterns. It compresses all detector rows on top of each other so if you want or need to see the row data detail, you must use Scan Analysis.

### Shorted channels:

Look at the pattern on the Modules and see if you have adjacent or near adjacent pins out of spec. The values are not guaranteed to be out of spec so look at the QIF tool for magnitudes, not just “red” or “green” indication. Use Scan Analysis to see the channel value details on a detector row. One pin very high and one adjacent pin very low (sometimes pinned to the extremes but not always) is the signature of a short between pins. A short to ground will have the offsets for only one channel at zero.

If you don’t see a pattern on the Modules, inspect the Convertor card layout on the Visualization tool and look for adjacent or near adjacent pins such that one pin is very high and one adjacent pin very low (sometimes pinned to the extremes but not always) is the signature of a short between pins. Follow the Convertor card problem troubleshooting tips.

Once you have established the problem is on the DAS-Detector Interface (DDIF), you must narrow it down to the backplane, backplane BGA connector, interposer, Detector BGA connector or the Detector. The easiest thing to do is to blow out the connection area with AeroDuster spray. Try this and re-scan.

If blowing out the connection area doesn’t work, the second easiest thing to do is to remove the flex. Scan using the Offset/Noise test and see if the short still exists. (Secure the flex if rotation is required.) If it is clear, look at the Detector flex connector for contaminants or metal flakes and correct the problem. Amax can be used, but make sure it is completely dry before reconnecting. Re-attach and re-scan. If the short re-appears, swap the flex with an adjacent interposer and see if it follows the flex. If the problem moves with the flex then the issue is with the detector. You must remove the “B” side flex and the dust shield before you remove the interposer.



### NOTICE

**Do not connect detector flexes from the “A” side to the “B” side or the other way around. You will damage the detector FETs and/or backplane requiring replacement of both.**

If the short stayed when the flex was removed, the fault is with the Interposer, backplane BGA or the backplane. Replace the interposer and re-scan. (You must remove the “B” side flex and the dust shield before you remove the interposer).

If the short is still there, the problem is with the backplane or the backplane connectors (192 pin convertor or 150 pin BGA). Inspect the backplane connectors under a bright light to look for contamination. If the “short” is near zero ohms, look for bent pins or metal flakes. If the “short” is in the 100 ohm to 10 k range, look for contamination. This is referenced to the Analog ground on the DAS backplane. Clean it and re-scan.

### Open Channels:

These can be found using DASTOOLS interconnect test. They are most likely open BGA connections or breaks on the Detector flexes. Sometimes they have been intermittent connections on the BGA-interposer interface.

Identify the suspect Convertor board slot and BGA location.

Swap the Convertor board with another board and re-scan. If it follows the board, replace the board.

Remove the Detector flex at the suspect location. Check the interposer for bent pins on the top. Try re-seating the flex connector and re-test. If that doesn’t work, swap the flex with an adjacent interposer and see if it follows the flex. If the problem moves with the flex then the issue is with the detector.

If the open does not follow the flex, the problem is with the interposer, backplane BGA connector or the 192 pin backplane Converter connector.

Remove the interposer and inspect it for bent pins both top and bottom. Replace as necessary.

Look at the DDIF backplane BGA connector for contamination.

Use a DVM and measure the resistance from the DDIF backplane pin to analog ground.

- Approximately 5 meg ohm = Backplane OK (Converter card path)
- Infinity = Open between DDIF and Converter
- All other = go to Shorted Channels on the previous pages.

If it is open, replace the backplane. If it isn't, re-assemble the DDIF and re-scan.

### 6.5.3 Interposer Issues

#### TROUBLESHOOTING -5V SHORTS TO AGND DUE TO BENT INTERPOSER PINS

-5V\_A shorts to AGND will show up as a group of cards or modules having the wrong FET modes. The MDAS16 backplanes have limiting resistors (392 ohms nominal) between the -5V\_A supply and the Detector flex module connections to prevent the backplane traces from fusing in the case of a short on the DDIF (DAS Detector InterFace) between the -5V\_A bias to the Detector FETs and AGND.

The backplanes are divided up into 4 groups of modules, each group having its own limiting resistor and test point. This is to make finding the shorted module easier.

|         | RIGHT<br>MODULES |         | CENTER<br>MODULES |         | LEFT<br>MODULES |         |
|---------|------------------|---------|-------------------|---------|-----------------|---------|
|         | A & B Sides      |         | A & B Sides       |         |                 |         |
| Group 1 | TP4              | 1 - 4   | TP9               | 19 - 23 | TP4             | 39 - 43 |
| Group 2 | TP3              | 5 - 9   | TP10              | 24 - 28 | TP3             | 44 - 48 |
| Group 3 | TP2              | 10 - 14 | TP11              | 29 - 33 | TP2             | 49 - 53 |
| Group 4 | TP1              | 15 - 18 | TP12              | 34 - 38 | TP1             | 54 - 57 |

**Table 2-7 DAS Backplane DVSS (-5V\_A) Test points**

You can determine whether there is a short between -5V\_A and AGND on the DDIF by measuring the impedance to ground at each test point with the power supplies off or you can measure the voltage relative to AGND with the supplies on.

The test point will read zero volts or ohms for a shorted module either way and you can use the table to determine which section has the -5V\_A short to AGND.

Once the module set is identified, leave the meter hooked to the test point and remove the Detector flexes from the "A" side, one at a time. When the short is cleared, the voltage at the test point will change to about -5V or the impedance will rise up over 300 ohms (typically).

If this does not happen, then remove the "B" side Detector flexes. If short is still present then remove the interposers until the short is cleared. Inspect both sides for bent pins.



#### NOTICE

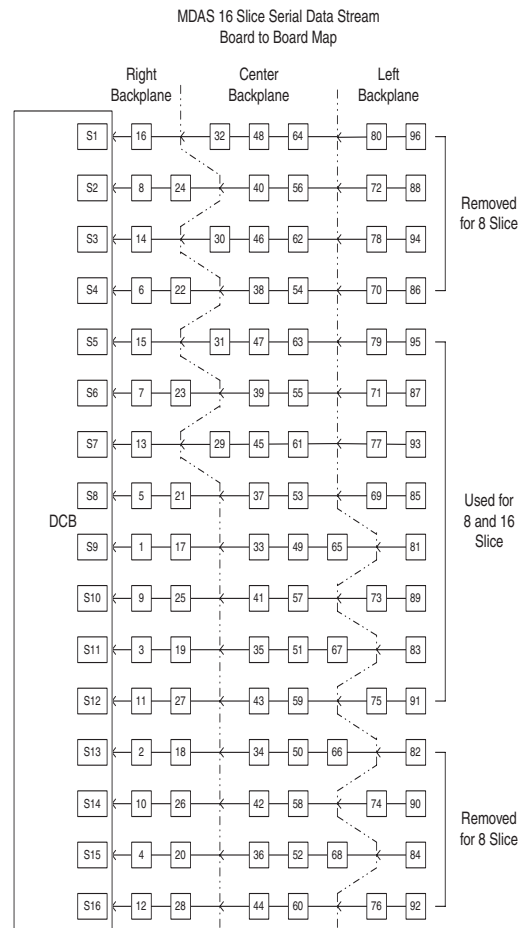
**Make sure you remove power as needed and use proper ESD protection to prevent additional damage.**

If you have done all of the above and the test point still reads zero, inspect the backplane and connectors for particles around the shorted module connector.

## 6.5.4 DAS Parity Errors

Data is sent from the high channel boards towards the DCB in 16 “serial streams” which consist of 2 sets of signals: odd and even. The “odd” set goes to pins B2 and B3 of the Converter cards and the “even” set goes to pins D2 and D3. If you are at this level of detail, you have big problems and must use the connector tables to troubleshoot pin/stream connections. Reference the Advanced Service Manual, “Jumpers, Switches, Adjustments, LEDs & Connections” for these detailed pinout tables.

For more general troubleshooting of parity errors, the serial stream chart will help you decipher where the problem is occurring. The software reports the first board in a stream that is causing errors. It may also report a single board with configuration problems. Attack that board first. If you have a single error listing, the boards with the problem may be the board noted or the board upstream depending if you have a receiver or transmitter problem. You could also have a cable problem. The serial stream chart will tell you if the error lies on a backplane (cable) boundary. If you have multiple streams with errors, the system error messages are misleading. It will tell you the same first board location for all streams. This means that some streams will report the wrong boards. This occurs because the firmware cannot distinguish multiple faults. Work the problem from stream 1 to 16 in order. Also start with the convertor nearest the DCB in the lowest failed stream.



**Figure 2-19 Serial Stream**

Example: Board 32 has a transmitter problem; Board 51 has a receiver problem. The system will report an error on boards [16 and/or 32] and [19 and/or 35] in streams 1 and 11 respectively. (NOTE WHAT IT REALLY SAYS FOR THE STREAM NAMES). See [Figure 1-7 on page 13](#) or [Figure 2-19](#).

*Logic tells us that we have at least 2 bad boards. Why?*

- *Because 2 different serial streams are being reported.*

- 1.) Start with the first or lowest serial stream reported and board nearest the DCB.
- 2.) Move board 16 to the far right of its stream (slot 96).
  - a.) The Error should now list [64 and/or 80] and [51 and/or 67].
  - b.) If not then return boards to the original slots and proceed.
- 3.) Move board 32 to the far right of its stream (slot 96).

The Error should now list [64 and/or 80] and [51 and/or 67].

*Since the error changed with the board movement, replace the board that is now in slot 96.*

*You still have a problem. The new error reports boards [64 and/or 80] and [51 and/or 67].*

*Applying logic to what you know.*

- *Original problem changed when board 32 was swapped with 96. Board 32 was bad.*
- *Bad board is now in slot 96, but board 96 is not listed in error log. Why?*
- *This means that the second bad board in the DAS is closer to the DCB than the bad board in slot 96. Troubleshoot the next serial stream listed. Serial stream 11 in this example.*

- 4.) Move board 51 to the far right of it's stream (slot 83).

The error should change to [80 and/or 96].

*Why only 2 possible boards being reported this time?*

- *Because the board in slot 96 has a bad transmitter and cannot send its data. The bad board in slot 83 has a bad receiver and might not report any error at all.*

Note: Do not attempt to "hide" bad boards in this fashion. They will fail causing you more grief and expense. Replace all bad boards when identified.

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### **Summary**

Any time you get an parity error and the boards are identified, you will not know for sure whether it is telling you the correct board, even if you only have one bad board. In that case, the board downstream of the indicated board could be bad (have a bad transmitter). Also chassis to chassis cabling and associated backplane connectors can cause this same issue. In this case modify the above procedure by swapping cards and or cables at the serial stream chassis breaks.







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